

CAU-Fusion: A Framework for Quantifying Confidence, Agreement and Uncertainty in Predictive Fusion

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A Capstone project report submitted in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering



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Declaration

We, Md Shadman Shakib Dip, Shahedur Rahman Rafin, Md. Moon Rahman Nayem, and Israt Jahan Tasnim hereby declare that the work presented in this capstone project report is the outcome of the investigation performed by us under the supervision of Supervisor Name, Designation, Department of Computer Science and Engineering, East West University. I/We also declare that no part of this project has been or is being submitted elsewhere for the award of any degree or diploma, except for publication.

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Letter of Acceptance

The capstone project report entitled "CAU-Fusion: A Framework for Quantifying Confidence, Agreement, and Uncertainty in Predictive Fusion" is submitted by Md Shadman Shakib Dip, Shahedur Rahman Rafin, Md. Moon Rahman Nayem, and Israt Jahan Tasnim to the Department of Computer Science and Engineering, East West University, Dhaka, Bangladesh, is accepted for the partial fulfillment of the requirement for the degree of Bachelor of Science in Computer Science and Engineering on 13/08/2026.

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Abstract

Reliable disease diagnosis often requires combining clinical biomarkers with medical imaging, yet most existing multi-modal systems fuse these two sources through simple averaging, an approach that ignores individual model confidence and inter-modality disagreement, often producing misleadingly confident results when the two sources conflict. To address this limitation, this work proposes CAU-Fusion, a late-fusion framework that combines clinical and imaging predictions using explicit quantification of Confidence, Agreement, and Uncertainty, rather than naive averaging. Unimodal posterior probabilities are first calibrated through a logit-sigmoid transformation and then combined via a closed-form fusion equation, in which a confidence term (cf), derived from prediction magnitude and inter-modal agreement, multiplicatively boosts the fused score, while a disagreement term (D) and a Shannon-entropy-based uncertainty term (U) act as dynamic penalty terms in the denominator, with their relative weights (α, β, γ) tuned through grid-search optimization. This design ensures that the fused confidence scales with the actual reliability of the underlying evidence: the proposed model is shown to avoid the overconfident and misleading predictions produced by simple averaging under conflicting evidence, instead flagging such cases as low-confidence for manual review. The framework is demonstrated by fusing a CatBoost clinical classifier with a DenseNet121 imaging model for diabetic retinopathy risk assessment, and is deployed as an interactive diagnostic tool that reports Low/Medium/High risk levels alongside modality-level explanations, with the underlying approach being directly generalizable to other diagnostic tasks that combine clinical and imaging modalities.

Keywords: Decision-level fusion, confidence-aware prediction, uncertainty quantification, disagreement modeling, diabetic retinopathy.

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1.1 Background

Medical diagnosis often requires information from multiple sources to provide reliable and comprehensive clinical decisions. While structured clinical data, such as laboratory test results, physiological measurements, and patient records, contain valuable information about a patient's health condition, medical images provide complementary visual evidence of disease characteristics that cannot be obtained from clinical data alone. Therefore, relying on a single data modality may lead to incomplete diagnostic information, motivating the adoption of multimodal predictive approaches that integrate heterogeneous data sources to improve diagnostic reliability [1], [2].

Recent advances in predictive modeling have enabled different machine learning techniques to specialize in different data modalities. Machine learning algorithms, such as CatBoost, are highly effective in analyzing structured clinical and tabular data, whereas deep learning architectures, such as DenseNet121, are capable of extracting discriminative features from medical images. Since these predictive models learn different types of information from different modalities, combining their prediction outputs has the potential to generate more complete and reliable diagnostic decisions than using either predictive model independently.

Consequently, multimodal predictive systems have received increasing attention in recent years because they exploit complementary information obtained from different data sources. Instead of depending solely on clinical measurements or medical images, these systems combine the strengths of multiple predictive models to improve overall prediction performance. In theory, integrating predictions from different modalities should produce more robust and dependable decisions, particularly for complex medical diagnosis tasks where no single modality can provide sufficient diagnostic evidence.

Despite these advantages, the predictive fusion process remains one of the major limitations of existing multimodal prediction systems. Most current approaches combine the outputs of independent predictive models using simple probability averaging, weighted averaging, or conventional ensemble techniques. These fusion methods assume that every predictive model contributes equally to the final decision regardless of the prediction context. Consequently, the fusion process does not evaluate whether an individual predictive model is highly confident in its prediction, whether different predictive models produce consistent or conflicting outputs, or whether uncertainty exists in the generated prediction. As a result, the final fused prediction may become overconfident even when the underlying predictive models disagree with one another, reducing the reliability and trustworthiness of AI-assisted decision support systems.

To overcome these limitations, there is a growing need for a general-purpose predictive fusion framework capable of evaluating the reliability of individual predictive models before generating the final decision. Rather than relying solely on conventional probability averaging, an effective fusion strategy should explicitly quantify prediction confidence, measure inter-model agreement, and estimate predictive uncertainty so that the fusion process becomes more cautious whenever conflicting evidence exists between different modalities. Such a reliability-aware predictive fusion framework can produce more accurate, interpretable, and trustworthy decisions and can be applied to a wide range of multimodal medical diagnosis problems involving both structured clinical data and medical images.

1.2 Problem Statement and Analysis

Many medical diseases require information from multiple diagnostic sources for reliable diagnosis. Clinical data provide valuable physiological and laboratory information, whereas medical images capture visual pathological characteristics that are often unavailable from structured clinical records. Consequently, multimodal prediction has become an important research direction because integrating these complementary sources of information can produce more reliable diagnostic decisions than relying on a single modality. However, the predictive fusion mechanism used in most existing multimodal systems remains limited, reducing the overall trustworthiness of the final prediction.

1.2.1 Problem Identification and Formulation

The primary problems addressed in this research are summarized as follows:

1. Dependence on Single-Modality Information: Many diseases cannot be diagnosed reliably using only structured clinical data or only medical images, since each modality provides only partial diagnostic evidence.

2. Conventional Predictive Fusion: Most multimodal systems combine independent clinical and image-based predictions using simple probability averaging, weighted averaging, or conventional ensemble techniques, assuming equal contribution from each predictive model regardless of its reliability.

3. Absence of Reliability Estimation: Existing predictive fusion methods generally ignore three important characteristics of predictive models: prediction confidence, inter-model agreement, and predictive uncertainty. Consequently, conflicting predictions from different modalities may still produce highly confident yet unreliable final decisions.

4. Limited Interpretability and Trustworthiness: The lack of reliability-aware predictive fusion reduces the transparency and interpretability of AI-assisted decision-support systems, making it difficult for users to assess the trustworthiness of the final prediction.

The figure 1 summarizes the key limitations of conventional predictive fusion, including single-modality dependence, equal contribution of model predictions, lack of reliability estimation, and limited interpretability. It further illustrates the need for CAU-Fusion to integrate confidence, agreement, and uncertainty for more reliable multimodal predictions.

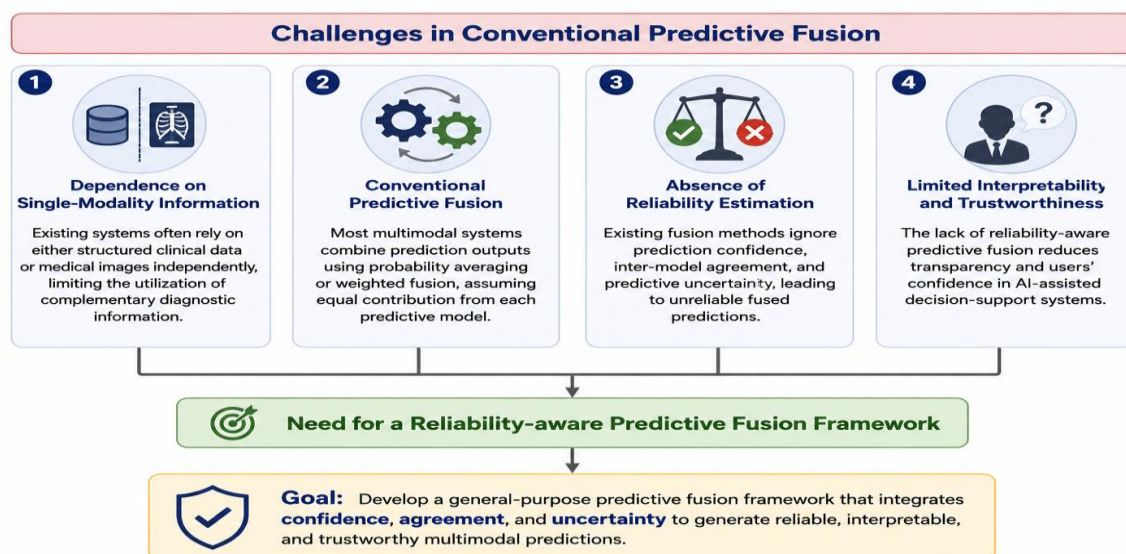


Figure 1: Problem Identification and Motivation for the Proposed CAU-Fusion Framework

1.2.2 Literature Review and Gap Analysis

Recent studies have demonstrated that combining structured clinical information with medical images generally improves predictive performance compared with single-modality approaches. Machine learning models are commonly employed for structured clinical data, while deep learning models are used for medical image analysis. Although multimodal prediction has become increasingly popular across various medical applications, most existing research primarily focuses on improving predictive accuracy rather than improving the reliability of the predictive fusion process. Most existing studies also give limited attention to prediction reliability and interpretability, making trustworthy predictive fusion an important research challenge.

The major research gap is that conventional predictive fusion strategies still rely on simple probability averaging or ensemble methods without evaluating the confidence of individual predictions, the agreement between different predictive models, or the uncertainty associated with the fused decision. Consequently, existing systems may generate misleadingly confident predictions when different modalities provide conflicting evidence. Moreover, these fusion methods treat all prediction outputs equally regardless of their reliability, reducing the robustness of the final decision. This limitation motivates the development of a more intelligent predictive fusion framework capable of producing reliable and explainable multimodal predictions

1.2.3 Problem Analysis

Based on the identified limitations and literature review, the following problem analysis is formulated:

- 1. Independent Prediction Generation:** Existing multimodal systems typically generate predictions independently from structured clinical data and medical images before combining them, preventing effective interaction between complementary diagnostic information during the prediction process.
- 2. Reliability-unaware Predictive Fusion:** Conventional predictive fusion methods treat prediction probabilities as equally reliable and perform simple probability averaging or weighted fusion without assessing the quality of individual predictions.
- 3. Lack of Confidence, Agreement, and Uncertainty Evaluation:** Most existing fusion frameworks do not explicitly evaluate prediction confidence, inter-model agreement, and predictive uncertainty before generating the final prediction. Consequently, conflicting predictions may still result in overconfident and unreliable decisions.
- 4. Limited Trustworthiness of Final Prediction:** Since the reliability of individual predictive models is not considered during fusion, the final prediction may lack transparency, robustness, and interpretability, reducing users' confidence in AI-assisted decision-support systems.
- 5. Need for a Reliability-aware Predictive Fusion Framework:** To overcome these limitations, a predictive fusion mechanism should evaluate the reliability of each predictive model prior to fusion by incorporating confidence estimation, agreement measurement, and uncertainty quantification. Such a framework can generate more trustworthy and interpretable predictions while remaining applicable to various multimodal medical prediction tasks.

6. **Application to Diabetic Retinopathy Prediction:** Although the proposed fusion framework is designed as a general-purpose multimodal predictive fusion strategy, **diabetic retinopathy** is selected as the case study because it naturally combines structured clinical information with retinal fundus images, providing an appropriate environment for validating the effectiveness of the proposed CAU-Fusion framework.

1.3 Project Objectives

The primary objective of this research is to develop CAU-Fusion, a general-purpose reliability-aware predictive fusion framework that improves the trustworthiness of multimodal medical prediction by incorporating prediction confidence, inter-model agreement, and predictive uncertainty into the predictive fusion process. Unlike conventional fusion techniques that rely on simple probability averaging, the proposed framework aims to generate more reliable, interpretable, and trustworthy predictions by evaluating the reliability of individual predictive models before producing the final decision.

The specific objectives of this research are as follows:

1. **Development of the CAU-Fusion Framework:** To design and develop a general-purpose predictive fusion framework that combines clinical and image-based model predictions using confidence, agreement, and uncertainty, instead of conventional probability averaging.
2. **Reliability-aware Predictive Fusion:** To develop a fusion mechanism that automatically becomes more cautious when different predictive models generate conflicting outputs, thereby improving the reliability of the final prediction.
3. **Explainable Decision Fusion:** To make the final fused prediction more interpretable by clearly indicating the contribution of each predictive model together with the associated confidence, agreement, and uncertainty information.
4. **General-purpose Applicability:** To develop a predictive fusion framework that is applicable to a broad range of multimodal medical prediction tasks involving structured clinical data and medical images.
5. **Validation through Diabetic Retinopathy:** To evaluate the effectiveness of the proposed CAU-Fusion framework using diabetic retinopathy as a real-world case study by integrating structured clinical information with retinal fundus images and comparing its performance with conventional predictive fusion methods.

Although the proposed CAU-Fusion framework is designed as a general-purpose reliability-aware predictive fusion framework, diabetic retinopathy (DR) is selected as the validation case study because it naturally combines structured clinical data and retinal fundus images, making it well suited for multimodal predictive fusion. Furthermore, according to the International Diabetes Federation (IDF) Diabetes Atlas, 11th Edition (2025), approximately 589 million adults worldwide are living with diabetes, and this number is expected to reach 853 million by 2050. Nearly 23% of diabetic patients develop diabetic retinopathy, while 11% suffer from sight-threatening diabetic retinopathy (STDR) [3], [4]. These statistics highlight the urgent need for reliable multimodal predictive frameworks for early screening and clinical decision-making, making diabetic retinopathy an appropriate real-world case study for validating the proposed CAU-Fusion framework. The figure 2 provides a visual overview of the relationship between the global diabetes burden and diabetic retinopathy, highlighting the progression from the overall diabetic population to undiagnosed diabetes and DR-related complications.

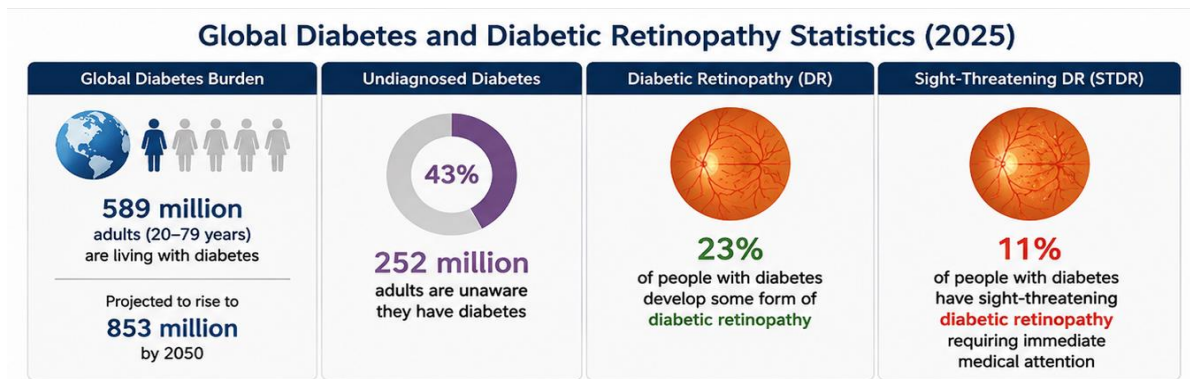


Figure 2: Global Diabetes Statistics Supporting the Selection of Diabetic Retinopathy (DR)

2.1 Survey of the State-of-the-Art

Diabetic retinopathy (DR) is one of the leading causes of vision impairment among individuals with diabetes, making early and accurate diagnosis an important research focus. Numerous studies have been conducted using retinal fundus images, structured clinical data, and multimodal predictive fusion to improve the detection and prediction of diabetic retinopathy. Recent research has also investigated confidence estimation, uncertainty quantification, and multimodal fusion strategies to enhance the reliability of predictive systems.

This section reviews previous studies on fusion-based predictive frameworks, multimodal diabetic retinopathy prediction, clinical data-based diabetes prediction, and retinal image analysis. The methodologies, datasets, experimental performance, strengths, and limitations of these studies are analyzed to identify the existing research gap and justify the proposed CAU-Fusion framework.

The study by He and Li [1] proposed an uncertainty-aware evidential fusion framework for semi-supervised medical image segmentation using evidential deep learning. The framework refines confidence and uncertainty estimation through evidential fusion and voxel-level asymptotic learning. It was evaluated on the LA, Pancreas-CT, ACDC, and TBAD datasets and achieved superior performance over existing methods, with a maximum Dice score of 92.93%. However, the proposed method is limited to medical image segmentation and does not address multimodal disease prediction.

Huang et al. [2] proposed a deep evidential fusion framework for multimodal medical image segmentation using uncertainty quantification and contextual discounting. The framework was evaluated on the BraTS2021 and HECKTOR datasets and outperformed existing state-of-the-art methods in segmentation performance and prediction reliability. However, the proposed framework is limited to medical image segmentation and does not address multimodal disease prediction using structured clinical and retinal image data.

Zou et al. [3] proposed EyeMoSt+, a confidence-aware multimodality learning framework for eye disease screening using Fundus and OCT images. The model employs Normal-Inverse Gamma (NIG) distributions and Mixture of Student's t (MoSt) fusion for uncertainty-aware prediction. It was evaluated on GAMMA, OLIVES, and an in-house dataset, achieving 81.4% accuracy under missing fundus and 67.29% accuracy under missing OCT conditions, outperforming existing multimodal methods. However, the framework requires both imaging modalities and does not utilize structured clinical data.

Mukherjee et al. [4] proposed an uncertainty-aware multimodal fusion framework for oral lesion classification using RGB images, hyperspectral features, and demographic data. The model employed ConvNeXt-v2 and an Incremental Heuristic Meta-Learner (IHML), achieving 64.56% accuracy and 66.23% Macro F1-score. However, the framework was evaluated only on oral lesion datasets.

According to the study of Nneji, G. U., et al. [5], a Weighted Fusion Deep Learning Network (WFDLN) was proposed for diabetic retinopathy detection using dual-channel fundus images. The model combines fine-tuned Inception V3 and VGG-16 with weighted feature fusion to improve classification performance. It achieved 98.5% accuracy on the Messidor dataset and 98.0% accuracy on the Kaggle dataset. However, the weighted fusion approach increases computational time and its performance may vary across datasets with different image quality and distributions.

Ramasamy et al. [6] proposed a feature fusion-based diabetic retinopathy detection method by combining GLCM, GLRLM, and Ridgelet features with an SMO classifier. The model was evaluated on the DIARETDB1 and Kaggle datasets, achieving 97.05% and 91.0% accuracy, respectively. However, the method relies on handcrafted feature extraction and may require higher computation time due to the large number of fused features.

Wardhani et al. [7] presented a comprehensive review of deep learning-based multimodal data fusion for diabetic retinopathy detection. The study analyzed different fusion strategies, modality combinations (Fundus, OCT, and EHR), and deep learning architectures, highlighting that multimodal approaches improve diagnostic performance. However, challenges such as limited standardized multimodal datasets, data heterogeneity, and poor model interpretability remain unresolved.

Li et al. [8] proposed a multimodal information fusion framework for proliferative diabetic retinopathy detection by combining structural OCT, OCT angiography (OCTA), and Line Scanning Ophthalmoscope (LSO) images. They evaluated early, intermediate, and hierarchical fusion strategies and found that hierarchical fusion achieved the best performance (AUC = 0.911), outperforming single-modality models. However, the study was conducted on a small dataset, indicating the need for larger-scale validation before clinical deployment.

Park et al. [9] developed an ensemble model by combining DNN, LightGBM, and XGBoost using 5,145 patient records and 86 laboratory features for disease prediction. The proposed model achieved 92% prediction accuracy and 81% F1-score, demonstrating the effectiveness of structured clinical data. However, the study focused only on laboratory test data and did not incorporate medical images or multimodal fusion.

According to the study of Afolabi et al. [10], diabetes prediction was performed using three supervised machine learning algorithms: Logistic Regression, Random Forest, and K-Nearest Neighbors (KNN) on electronic health records (EHR). Among the evaluated models, KNN achieved the highest accuracy of 96.09%, with 98.54% sensitivity and 93.63% specificity, outperforming the other models. However, the study considered only traditional machine learning algorithms and did not investigate advanced deep learning approaches.

Cervantes et al. [11] presented a comprehensive survey of Support Vector Machine (SVM) classification, highlighting its effectiveness in handling high-dimensional data and binary classification tasks. The study discussed SVM applications, challenges, and recent trends, concluding that SVM is a reliable machine learning algorithm for medical diagnosis, including diabetes prediction.

Shin et al. [12] developed multiple machine learning-based diabetes prediction models using electronic health records with Gradient Boosting and Random Forest algorithms. The best model achieved an ROC-AUC of 0.928, demonstrating high predictive performance using routine health screening data, though external validation was still required.

Hasan et al. [14] proposed a machine learning-based framework for early-stage diabetes prediction using the UCI Early-Stage Diabetes Risk Prediction dataset. The study applied feature transformation techniques with multiple classifiers and feature selection methods. The Extra Trees classifier with StandardScaler and PCC achieved the best performance, reaching 99.93% ROC-AUC, 99.36% F1-score, 99.72% precision, and 99.01% recall. However, the model was evaluated on a single structured tabular dataset, limiting its generalizability to diverse real-world clinical data.

According to the study of Rezaei et al. [15], an automated multimodal framework was proposed for diabetic retinopathy (DR) severity assessment by combining Ultra-Widefield Color Fundus (UWF-CFP) images with 76 clinical tabular features using an MLP-ResNet50 fusion model. The model was evaluated on the EVIRED dataset and achieved 63.4% multiclass accuracy with a Kappa score of 0.807, outperforming the image-only model by 2.1% accuracy. The framework was further validated on the BRset and mBRset datasets, where multimodal fusion consistently improved classification performance over unimodal methods.

However, the study reported limited performance for the PDR class due to the small number of PDR samples in the primary dataset.

Oulhadj et al. [16] proposed a hybrid deep learning model that combines a modified Inception block, Capsule Network, and Discrete Wavelet Transform (DWT) pyramid hierarchy for diabetic retinopathy detection. The model improved feature extraction by capturing both multiscale image details and spatial relationships, achieving a testing accuracy of 86.54%.

2.2 Related Works Table

This Table 1 presents a comparative review of previous studies related to multimodal medical prediction, uncertainty-aware fusion, machine learning, and diabetic retinopathy and diabetes detection. It compares the studies based on their methodology, dataset, reported performance, and key strengths or limitations to identify the research gap addressed by the proposed CAU-Fusion framework.

Table 1: Summary of Related Works

Serial No.	Paper Title	Methodology	Dataset	Accuracy	Key Strengths & Weaknesses
1	Uncertainty-aware Evidential Fusion-based Learning for Semi-supervised Medical Image Segmentation	Evidential Deep Learning + Uncertainty Fusion	LA, Pancreas-CT, ACDC, TBAD	Dice $\uparrow$ (Best among compared methods)	Uncertainty-aware fusion improves segmentation / No DR application
2	Deep Evidential Fusion with Uncertainty Quantification and Contextual Discounting	Deep Evidential Fusion	BraTS, MMWHS	Dice $\uparrow$ (Improved over baseline)	Reliable uncertainty estimation / Computationally complex
3	Confidence-aware Multi-modality Learning for Eye Disease Screening	EyeMoSt+, NIG + MoSt Fusion	GAMMA, OLIVES, In-house	82.9%	Robust to missing modalities / Requires multimodal images
4	Patient-Aware Multimodal RGB-HSI Fusion via Incremental Heuristic Meta-Learning	ConvNeXt-v2 + IHML Fusion	Oral lesion dataset	64.56%	Multimodal uncertainty fusion / Limited to oral lesions

5	Identification of Diabetic Retinopathy Using Weighted Fusion Deep Learning Based on Dual-Channel Fundus Scans	Weighted Fusion (InceptionV3 + VGG16)	Messidor, Kaggle	98.5%, 98.0%	High DR accuracy / Higher computational cost
6	Detection of Diabetic Retinopathy Using Fusion of Textural and Ridgelet Features	GLCM + GLRLM + Ridgelet + SMO	DIARETDB1, Kaggle	97.05%, 91.0%	Effective handcrafted fusion / Feature engineering required
7	Deep Learning for Diabetic Retinopathy Detection: A Review of Multimodal Data Fusion Approaches	Literature Review	Multiple studies	Not applicable	Comprehensive survey / No experimental validation
8	Multimodal Information Fusion for the Diagnosis of Diabetic Retinopathy	Hierarchical Multimodal Fusion	Private DR dataset	AUC = 0.911	Multimodal fusion improves diagnosis / Small dataset
9	Development of Machine Learning Model for Diagnostic Disease Prediction Based on Laboratory Tests	DNN + LightGBM + XGBoost Ensemble	5,145 patients	92%	Strong laboratory-based prediction / No image modality

10	Predicting Diabetes Using Supervised Machine Learning Algorithms on E-Health Records	LR, RF, KNN	EHR dataset	96.09% (KNN)	High prediction accuracy / Only structured data
11	Diabetes Prediction Using Machine Learning and Explainable AI Techniques	ML + Explainable AI (XAI)	Not specified	Not reported	Interpretable prediction / Limited performance comparison
12	Development of Various Diabetes Prediction Models Using Machine Learning Techniques	Gradient Boosting, Random Forest	Health screening dataset	ROC-AUC = 0.928	High prediction performance / Requires external validation
13	Prediction and Diagnosis of Future Diabetes Risk	Machine Learning Classifiers	Pima Indians Diabetes Dataset	≈78%	Early diabetes risk prediction / Moderate accuracy
14	A Machine Learning-Based Model for Early Diabetes Detection	Extra Trees + Feature Selection	UCI Early-Stage Diabetes Dataset	99.93% ROC-AUC	Excellent prediction performance / Single tabular dataset
15	Automated Multimodal Severity Assessment of Diabetic Retinopathy	MLP + ResNet50 Fusion	EVIRET, BRset, mBRset	63.4%	Image + clinical data fusion / Poor PDR performance

16	Diabetic Retinopathy Detection Using DL-Based Feature Extraction and Image Processing Techniques	Modified Inception + Capsule Network + DWT	APTOS 2019	86.54%	Multiscale feature extraction / Moderate accuracy
17	A Deep Learning Based Model for Diabetic Retinopathy Grading	CNN-based DR Grading	APTOS 2019	97.83%	High grading accuracy / Single benchmark dataset
18	Automated Multidimensional Deep Learning Platform for Referable Diabetic Retinopathy Detection	Multidimensional CNN	Multi-center retinal dataset	AUC = 0.955	Large-scale validation / Focused on referable DR
19	Early Detection and Severity Classification of Diabetic Retinopathy Using CNN	Convolutional Neural Network	APTOS 2019	98.4%	High classification accuracy / Limited dataset diversity
20	An Advanced Deep Neural Network for Fundus Image Analysis	Deep Neural Network	APTOS 2019	98.3%	Robust fundus analysis / Image-only approach
21	Predicting diabetes using machine learning: a comparative study of classifiers	Neural Network	Pima Indians Diabetes Dataset	78.57 %	Comprehensive review of ML methods / No experimental validation

22	Diabetes Prediction and Management Using Machine Learning Approaches	Classifiers	Clinical Patient Dataset	99.79%	Discusses prediction and management strategies / Limited quantitative evaluation
23	An Intelligent Machine Learning Pipeline for Early Diabetes Prediction	Ensemble Machine Learning Pipeline	Early diabetes dataset	99.2%	High prediction accuracy / Limited external validation
24	Deep Learning for the Detection and Classification of Diabetic Retinopathy	CNN-based Deep Learning	Multiple public DR datasets	97.2%	Comprehensive DR classification / High computational cost
25	Diabetes Detection from Non-Diabetic Retinopathy Fundus Images Using Deep Learning	CNN-based Fundus Image Analysis	EyePACS	AUC = 0.90	Detects diabetes from retinal images / Requires large retinal image datasets

2.3 Summary

The literature review covers a wide range of studies on diabetic retinopathy prediction using retinal fundus images, structured clinical data, and multimodal learning approaches. Several studies employ machine learning, deep learning, and multimodal fusion techniques to improve diagnostic performance. Recent research also introduces confidence-aware and uncertainty-aware learning strategies to enhance prediction reliability. Overall, multimodal prediction has shown promising results by utilizing complementary information from different data sources and improving diagnostic accuracy. However, existing predictive fusion methods mainly rely on probability averaging, weighted averaging, or conventional ensemble techniques without adequately considering prediction confidence, inter-model agreement, and predictive uncertainty. These limitations may lead to unreliable predictions when different modalities provide conflicting evidence. Addressing these challenges, the proposed CAU-Fusion framework aims to develop a more reliable and interpretable multimodal predictive fusion strategy by explicitly incorporating confidence, agreement, and uncertainty into the final prediction process, thereby improving the trustworthiness of AI-assisted diabetic retinopathy prediction.

3.1 Materials

The proposed Confidence, Agreement, and Uncertainty-aware Fusion (CAU-Fusion) framework was developed through a systematic process involving dataset collection, dataset exploration, dataset sampling, data preprocessing, and research environment setup. First, clinical records and corresponding retinal fundus images were collected and linked using a unique Image_ID to establish a one-to-one correspondence. The datasets were then explored to examine their structure, characteristics, data quality, and image properties. For sampling, a stratified train-test split and SMOTETomek were applied to the clinical data, while the image data were divided into training and validation sets with data augmentation applied only to the training images. Clinical data were preprocessed through duplicate removal, inconsistent-value correction, missing-value handling, and encoding, while retinal images were resized to 224×224 pixels, normalized, and augmented. The research was implemented in Python using Kaggle Notebook, with Scikit-learn, TensorFlow, Keras, and supporting libraries.

3.1.1 Dataset Collection

To evaluate the proposed Confidence, Agreement, and Uncertainty-aware Fusion (CAU-Fusion) framework, a multimodal dataset comprising paired clinical records and retinal fundus images was utilized. The dataset consists of two complementary modalities: clinical (tabular) data and retinal fundus image data.

The clinical dataset includes demographic attributes (Age and Gender), renal function indicators (Urea and Creatinine), glycemic and metabolic markers (HbA1c and BMI), and a comprehensive lipid profile consisting of Cholesterol, Triglycerides (TG), HDL, LDL, and VLDL. The corresponding retinal fundus image dataset contains retinal images of the same patients, enabling visual assessment of diabetes-related retinal characteristics. The dataset contains records from both diabetic and non-diabetic patients.

To construct the multimodal dataset, each patient's clinical record was linked to its corresponding retinal fundus image using a unique Image_ID, establishing a strict one-to-one correspondence between the two data modalities. This pairing ensures that both the clinical information and retinal fundus image belong to the same patient, forming a unified multimodal dataset for the proposed CAU-Fusion framework. Figure 3 shows the one-to-one mapping between a patient's clinical record and the corresponding retinal fundus image using Image_ID. It illustrates how the clinical features and retinal image are paired for the same patient.

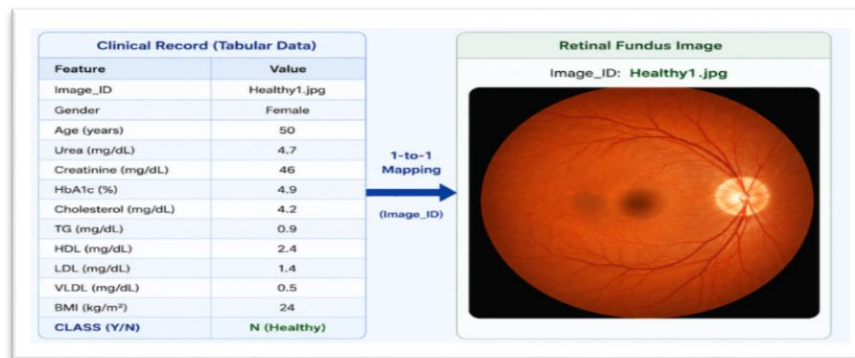


Figure 3: One-to-One Mapping Between a Clinical Record and Its Corresponding Retinal Image

3.1.2 Dataset Exploration

Following the dataset collection process, an exploratory analysis was conducted to examine the structure, characteristics, and overall quality of both the clinical and retinal fundus image datasets before the preprocessing stage. This analysis provided a comprehensive understanding of the available data and verified its suitability for the proposed multimodal learning framework.

For the clinical dataset, the exploration focused on identifying feature types, data distribution, and data quality. The dataset comprises both numerical and categorical attributes. The numerical features include Age, Urea, Creatinine, HbA1c, BMI, Cholesterol, Triglycerides (TG), HDL, LDL, and VLDL, whereas Gender represents the categorical feature. In addition, the target variable (CLASS) consists of two classes, namely Diabetic (Y) and Non-Diabetic (N), which serve as the ground-truth labels for binary classification. The dataset was further examined to identify missing values, duplicate records, feature completeness, and the consistency of the Image_ID field.

For the retinal fundus image dataset, exploratory analysis was performed to assess the overall image quality and dataset characteristics. The analysis included sample image visualization, image resolution inspection, brightness distribution, RGB color distribution, pixel intensity analysis, blur detection, duplicate and corrupted image identification, and dataset diversity assessment. These analyses ensured that the image dataset was suitable for subsequent preprocessing and deep learning model development.

The findings from both datasets established a reliable foundation for the subsequent data preprocessing and multimodal model development stages.

3.1.3 Dataset Sampling

To ensure reliable model training and unbiased performance evaluation, appropriate sampling strategies were applied separately to the clinical and retinal fundus image datasets according to their respective characteristics.

For the clinical dataset, a stratified train-test split was performed to preserve the original distribution of diabetic (Y) and non-diabetic (N) samples across the training and testing subsets. Since the dataset exhibited class imbalance, the SMOTETomek sampling technique was applied to the training data. SMOTETomek combines the Synthetic Minority Over-sampling Technique (SMOTE) with Tomek Links under-sampling to generate synthetic minority class samples while removing overlapping and noisy instances between classes. This strategy improves class balance and enhances the learning capability of the machine learning models.

For the retinal fundus image dataset, the images were divided into training and validation subsets prior to model development. Instead of applying resampling techniques, data augmentation was performed on the training images to increase data diversity and improve the generalization capability of the deep learning models. The validation dataset remained unchanged to provide an unbiased evaluation of model performance.

To maintain the integrity of the evaluation process, all sampling and augmentation procedures were applied only to the respective training datasets, while the testing and validation datasets were kept unchanged for objective performance assessment.

3.1.4 Data Preprocessing

Prior to model development, preprocessing techniques were applied separately to the clinical and retinal fundus image datasets to improve data quality and prepare the data for subsequent machine learning and deep learning models.

The clinical dataset was preprocessed by removing duplicate records and correcting inconsistent values. Missing values were handled using an appropriate imputation strategy to ensure data completeness. The categorical feature (Gender) and the target variable (CLASS) were encoded into numerical representations suitable for machine learning models. Finally, the processed dataset was organized into input features (X) and target labels (y) for model training.

The retinal fundus images were preprocessed by resizing all images to 224×224 pixels to maintain a uniform input size. The pixel values were normalized prior to training. To improve model robustness and reduce overfitting, data augmentation techniques, including rotation, zooming, and horizontal flipping, were applied to the training images. Finally, the processed images were organized into batches using ImageDataGenerator for efficient deep learning model training.

These preprocessing steps ensured that both the clinical and retinal fundus image datasets were prepared in a consistent manner, providing high-quality inputs for the subsequent multimodal learning framework.

3.1.5 Research Environment & Devices

The proposed Confidence, Agreement, and Uncertainty-aware Fusion (CAU-Fusion) framework was developed and evaluated using the Kaggle Notebook platform, which provides a cloud-based computational environment for machine learning and deep learning experiments. The implementation was carried out in Python using the Jupyter Notebook interface.

The research environment consisted of the following hardware and software resources:

- **Development Platform:** Kaggle Notebook (Jupyter Notebook environment)
- **Operating System:** Windows 10 (64-bit)
- **Local System:** Intel® Core™ i5-10300H CPU @ 2.50 GHz, 8 GB RAM, NVIDIA GeForce GTX 1650 (4 GB)
- **Cloud Computing Resource:** NVIDIA Tesla T4 GPU provided by Kaggle
- **Programming Language:** Python
- **Machine Learning Library:** Scikit-learn
- **Deep Learning Frameworks:** TensorFlow and Keras
- **Supporting Libraries:** Pandas, NumPy, OpenCV, Matplotlib, and Seaborn

The combination of local computing resources and Kaggle's cloud-based infrastructure provided an efficient environment for data preprocessing, model development, multimodal fusion, and performance evaluation.

3.2 Method

The proposed CAU-Fusion framework follows a multimodal methodology where clinical and retinal fundus image data are processed through separate pipelines. The selected CatBoost and DenseNet121 models generate the clinical probability (P_c) and image probability (P_i), respectively. These probabilities are then integrated through the CAU-Fusion framework using probability calibration, confidence estimation, agreement and disagreement analysis, and uncertainty estimation. Finally, the fusion components are combined through the proposed fusion function to generate the final probability (P_{final}) and prediction.

3.2.1 Methodology

The Confidence, Agreement, and Uncertainty-aware Fusion (CAU-Fusion) framework follows a multimodal methodology that integrates paired clinical records and corresponding retinal fundus images for diabetes prediction. Initially, the clinical data undergo dataset exploration, sampling, preprocessing, machine learning model development, and performance evaluation, through which the selected CatBoost model generates the clinical prediction probability (P_c). Similarly, the retinal fundus images undergo dataset exploration, sampling, preprocessing, data augmentation, CNN model development, and performance evaluation, through which the selected DenseNet121 model generates the image prediction probability (P_i). The prediction probabilities obtained from both modalities are then integrated through the CAU-Fusion framework, which performs probability calibration, feature engineering, hyperparameter tuning, threshold optimization, confidence estimation, agreement analysis, and uncertainty estimation. Finally, the fused probability is utilized to generate the final prediction. Figure 4 shows the overall methodology of the CAU-Fusion framework, showing the parallel clinical and retinal fundus image pipelines, followed by the integration of their prediction probabilities through the CAU-Fusion framework to generate the final prediction.

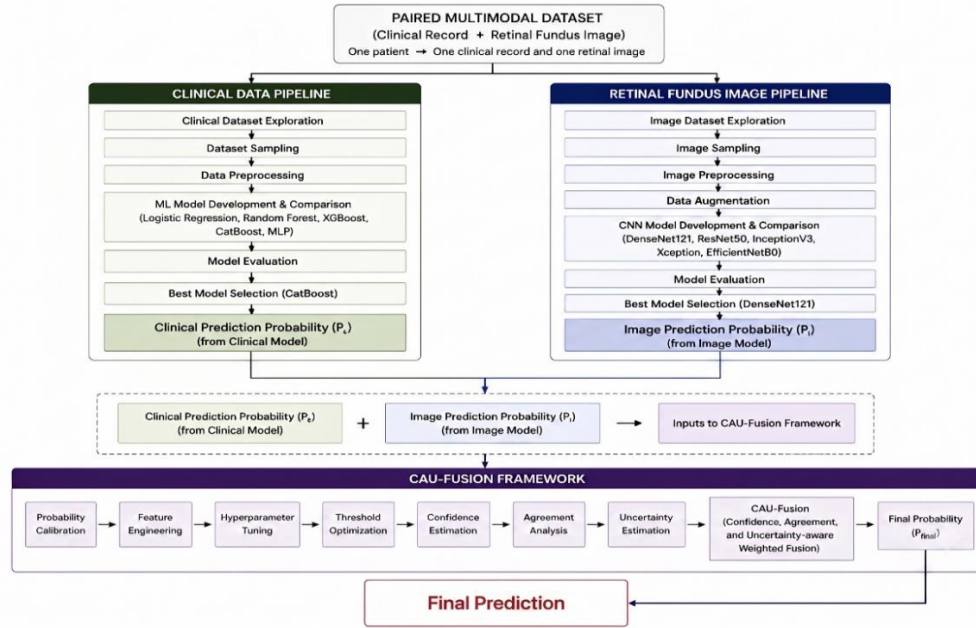


Figure 4: Proposed Methodology

3.2.2 Proposed Model

The proposed Confidence, Agreement, and Uncertainty-aware Fusion (CAU-Fusion) model is designed to enhance multimodal diabetes prediction by effectively integrating the prediction probabilities generated from the selected clinical and retinal fundus image models. The model receives the clinical prediction probability (P_c) from the CatBoost model and the image prediction probability (P_i) from the DenseNet121 model as its primary inputs. Initially, a base fusion score (S) is computed by combining the prediction probabilities from both modalities. The fusion process is subsequently refined by incorporating a confidence score (C_f) to strengthen reliable predictions, while disagreement (D) and uncertainty (U) are treated as penalty terms to reduce the influence of conflicting and ambiguous predictions. Finally, the proposed fusion function integrates these components to produce the final probability (P_{final}), which serves as the output of the proposed CAU-Fusion model.

3.3 Design / Framework

The proposed CAU-Fusion framework consists of three main components: CatBoost, DenseNet121, and the CAU-Fusion model. CatBoost is used to generate the clinical prediction probability (P_c), while DenseNet121 generates the retinal image prediction probability (P_i). These probabilities are then calibrated and combined through the CAU-Fusion framework using Simple Fusion, Agreement, Confidence, Disagreement, and Entropy-based Uncertainty. The final probability (P_{final}) is calculated using confidence-based boosting and disagreement and uncertainty penalties. Finally, α , β , γ , and the decision threshold are optimized through grid search to generate the final prediction.

3.3.1 CatBoost

CatBoost has been selected as the primary machine learning model for the clinical data due to its strong performance and efficiency in handling structured datasets. One of the main reasons for selecting CatBoost is its ability to process both categorical and numerical features effectively with minimal preprocessing. It employs an ordered boosting technique, which helps reduce overfitting and minimizes target leakage during training. Furthermore, CatBoost constructs symmetric decision trees, enabling faster computation and improved optimization compared to conventional gradient boosting algorithms. From an architectural perspective, CatBoost is based on the gradient boosting framework, where an ensemble of decision trees is built sequentially. Each successive tree learns from the residual errors of the previous trees, thereby improving the overall prediction performance. During training, the model iteratively minimizes the loss function and combines the outputs of multiple decision trees to generate the final prediction. This learning strategy allows CatBoost to effectively capture complex relationships within structured clinical data while maintaining high prediction stability. Based on the experimental evaluation, CatBoost demonstrated the best overall performance among the evaluated machine learning models and was therefore selected as the clinical prediction model for the proposed CAU-Fusion framework. Figure 5 shows the symmetric decision-tree structure used in CatBoost, where feature-based conditions divide the input data into different branches. Each decision path leads to a leaf node, which contributes to the final prediction.

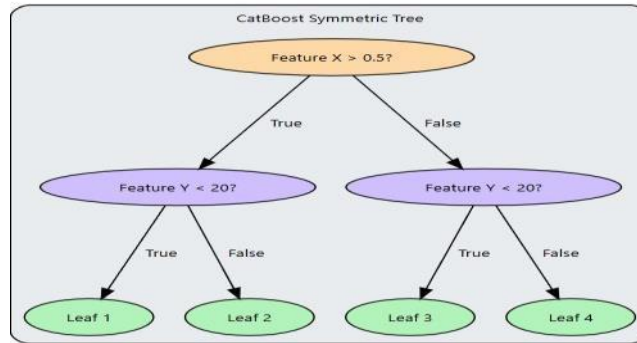


Figure 5: CatBoost Architecture

3.3.2 DenseNet121

DenseNet121 has been selected as the deep learning model for retinal fundus image analysis due to its high effectiveness in medical image classification tasks. One of its key advantages is its ability to overcome the vanishing gradient problem by establishing direct connections between all layers, thereby ensuring efficient information flow throughout the network. Furthermore, DenseNet121 promotes feature reuse, which reduces the number of parameters while maintaining high classification performance. From an architectural perspective, DenseNet121 consists of 121 layers organized into multiple dense blocks and transition layers. Each layer receives feature maps from all preceding layers within a dense block, which enhances feature propagation and encourages the reuse of learned representations. The network begins with an initial

convolutional layer, followed by a series of dense blocks interconnected by transition layers for feature compression and downsampling. Finally, global average pooling and a fully connected layer generate the image prediction probability (P_i), which is utilized by the proposed CAU-Fusion framework. Based on the experimental evaluation, DenseNet121 demonstrated the best overall performance among the evaluated convolutional neural network (CNN) models and was therefore selected as the retinal fundus image prediction model for the proposed CAU-Fusion framework. Figure 6 shows the DenseNet121 architecture, consisting of four dense blocks connected by transition layers. The network progressively extracts and reuses image features before applying the final pooling and classification layers to generate the image prediction.

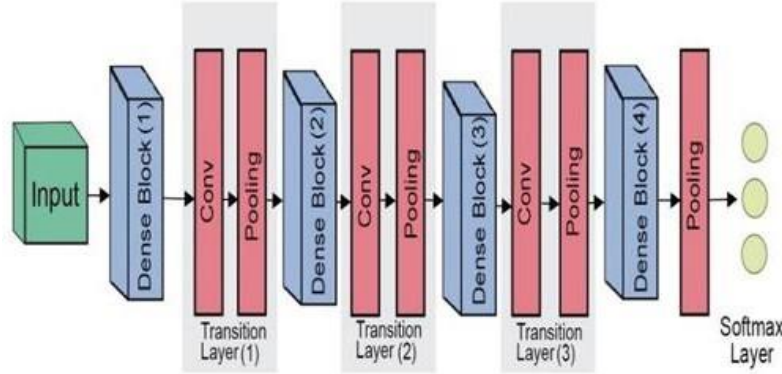


Figure 6: DenseNet121 Architecture

3.3.3 CAU-Fusion Framework

The proposed Confidence, Agreement, and Uncertainty-aware Fusion (CAU-Fusion) framework is designed to integrate the prediction probabilities generated from the selected clinical and retinal fundus image models for robust multimodal diabetes prediction. The framework receives the clinical prediction probability (P_c) from the CatBoost model and the image prediction probability (P_i) from the DenseNet121 model as its primary inputs. Initially, a base fusion score (S) is computed by combining the prediction probabilities from both modalities. To improve the reliability of the final prediction, the framework incorporates three complementary measures: confidence estimation, agreement analysis, and uncertainty estimation. The confidence score (C_f) strengthens reliable predictions, whereas disagreement (D) and uncertainty (U) are treated as penalty terms to reduce the influence of conflicting and ambiguous predictions. These measures enable the framework to adaptively adjust the contribution of each modality based on the quality and consistency of their predictions. Finally, the proposed CAU-Fusion strategy integrates the calibrated prediction probabilities, confidence score, disagreement, and uncertainty through the fusion function to generate the final probability (P_{final}). This probability serves as the basis for the final prediction, enabling a more balanced and reliable multimodal diabetes prediction compared with conventional probability fusion approaches. Figure 7 shows the CAU-Fusion mechanism, where the clinical prediction probability (P_c) and image prediction probability (P_i) are combined to obtain a base fusion score (S). The framework incorporates confidence as a positive factor and disagreement and uncertainty as penalty terms to generate the final prediction probability (P_{final}).

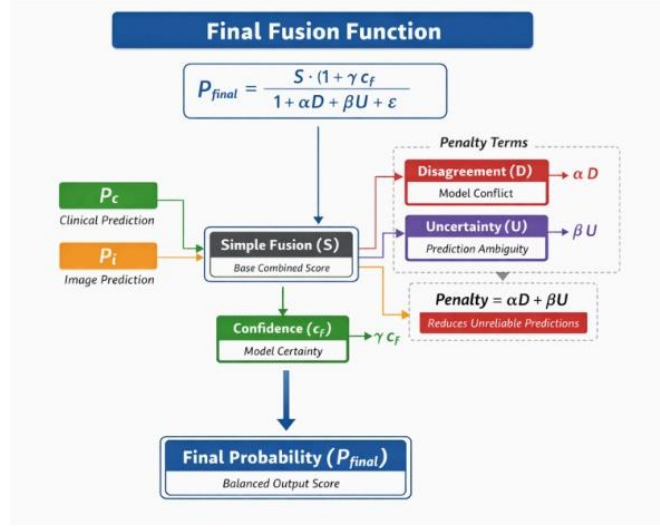


Figure 7: CAU-Fusion Framework

3.3.4 Algorithm / Model Formulation

The proposed Confidence, Agreement, and Uncertainty-aware Fusion (CAU-Fusion) framework integrates the prediction probabilities generated from the selected clinical and retinal fundus image models through a confidence-aware probability fusion strategy. Let P_c and P_i denote the prediction probabilities generated by the CatBoost and DenseNet121 models, respectively. Initially, these probabilities are combined to compute a base fusion score (S). The fusion process is subsequently refined by incorporating a confidence score (C_f), while disagreement (D) and uncertainty (U) are introduced as penalty terms to reduce the influence of conflicting and uncertain predictions. The final prediction probability is computed using the following formulation:

$$P_{final} = \frac{S(1 + \gamma C_f)}{1 + \alpha D + \beta U + \epsilon}$$

where P_{final} represents the final prediction probability generated by the proposed CAU-Fusion framework. S denotes the base fusion score computed from the clinical and image prediction probabilities, while C_f represents the confidence score that enhances reliable predictions. D indicates the disagreement score between the prediction probabilities obtained from the two modalities, and U represents the uncertainty score associated with the multimodal prediction. The parameters α and β control the contributions of the disagreement and uncertainty penalty terms, respectively, whereas γ determines the influence of the confidence score during the fusion process. Finally, ϵ is a small positive constant introduced to prevent division by zero.

The proposed formulation increases the contribution of reliable predictions through the confidence score while reducing the influence of conflicting and uncertain predictions using disagreement and uncertainty penalties. Consequently, the framework produces a balanced and robust final prediction probability for multimodal diabetes prediction.

System Overview

This report presents a late-stage probability fusion framework designed to combine output predictions from two distinct modalities: a clinical data model producing probability P_c , and a medical imaging model producing probability P_i . Traditional fusion strategies typically apply basic arithmetic averaging or fixed linear weighting across models. In contrast, the framework described here dynamically adjusts the final confidence level by evaluating prediction agreement and individual model uncertainty.

System Processing Pipeline

1. Extract unimodal probability scores from the clinical (P_c) and imaging (P_i) models.
2. Apply logit and sigmoid transformations for probability calibration.
3. Compute core metrics: Simple Fusion (S), Confidence (cf), Disagreement (D), and Uncertainty (U).
4. Calculate fused probabilities using parameterized penalty terms (α, β, γ).
5. Optimize decision thresholds to select the final classification output.

3.3.5 Mathematical Framework

Let P_c and P_i represent the raw output probabilities from the clinical and imaging classifiers respectively, where $P_c, P_i \in [0, 1]$.

3.3.6 Probability Calibration via Log-Odds Transformation

Raw model outputs often display calibration skew or overconfidence near extreme values. To address this, probabilities are mapped into unbounded log-odds space using the logit transformation:

$$\text{logit}(P) = \ln (P / (1 - P)), \quad P \in (0, 1) \quad (1)$$

To prevent asymptotic instability as P approaches 0 or 1, inputs are first clipped using a small clipping parameter $\varepsilon = 10^{-7}$:

$$\tilde{P} = \text{clip}(P, \varepsilon, 1 - \varepsilon) \quad (2)$$

The calibrated probabilities P'_c and P'_i are then reconstructed using the standard logistic sigmoid function σ :

$$P'_c = \sigma(\text{logit}(\tilde{P}_c)), \quad P'_i = \sigma(\text{logit}(\tilde{P}_i)) \quad (3)$$

3.3.7 Simple Fusion (S)

The simple fusion score S serves as the unweighted baseline, representing the unweighted mean of the calibrated predictions:

$$S = 0.5 \cdot P'_c + 0.5 \cdot P'_i \quad (4)$$

The value of S strictly resides in the interval $[0, 1]$ and provides a symmetric reference point, assuming both modalities contribute equally reliable signal.

3.3.8 Agreement Component

The agreement component quantifies the degree of consensus between the two modality-specific predictions. It is derived from the disagreement score D , defined as the absolute difference between the two calibrated probabilities:

$$D = |P'_c - P'_i|$$

Agreement is then expressed as the complement of disagreement:

$$\text{Agreement} = 1 - D$$

A value of $D = 0$ indicates perfect consensus between the clinical and imaging models, whereas $D = 1$ represents maximal disagreement (i.e., one model predicts near-certain positive while the other predicts near-certain negative). Agreement is subsequently used to gate the confidence component, ensuring that a confidence boost is applied only when the two modalities are in accord, rather than when either model is independently confident.

3.3.9 Confidence Component (cf)

The confidence component measures the joint reliability of the fused prediction by combining each model's individual certainty with the degree of inter-model agreement. Each model's confidence reflects how far its predicted probability deviates from the decision boundary (0.5):

$$conf_c = 2 \times |P'_c - 0.5|, \quad conf_i = 2 \times |P'_i - 0.5| \quad (5)$$

These individual confidences are averaged to obtain the raw confidence:

$$raw_confidence = (conf_c + conf_i) / 2 \quad (6)$$

The final confidence factor is then obtained by scaling this raw confidence with the agreement score:

$$cf = Agreement \times raw_confidence = (1 - D) \times raw_confidence \quad (7)$$

This agreement-gated formulation ensures that cf increases only when both modalities are simultaneously confident and in agreement. For instance, when $P_c = 0.95$ and $P_i = 0.90$, the two models are both highly confident and closely aligned, yielding cf close to 1.0 and thereby boosting the overall fused score. Conversely, if the two modalities diverge sharply — for example, $P_c = 0.95$ and $P_i = 0.10$ — the high individual confidences are suppressed by the low agreement term, preventing the model from producing an overconfident fused result from conflicting evidence. This behaviour distinguishes the proposed confidence formulation from a naive magnitude-based measure, as it explicitly penalizes inter-modal conflict rather than rewarding the stronger of the two predictions in isolation.

3.3.10 Disagreement Metric (D)

To quantify conflict between the clinical and imaging predictions, the disagreement metric calculates the absolute L1 distance between the two calibrated probabilities:

$$D = |P'_c - P'_i| \quad (8)$$

A value of $D = 0$ reflects perfect alignment between modalities, whereas values closer to 1.0 indicate strong inter-model conflict.

3.3.11 Entropy-Based Uncertainty (U)

Prediction ambiguity is evaluated using Shannon binary cross-entropy $H(p)$, which measures proximity to the decision boundary at $p = 0.5$:

$$H(p) = - [p \cdot \log_2(p) + (1 - p) \cdot \log_2(1 - p)] \quad (9)$$

$$U = (H(P'_c) + H(P'_i)) / 2 \quad (10)$$

Uncertainty U reaches its maximum value of 1.0 when predictions reside around 0.5, and drops to 0.0 when predictions are highly definitive (near 0 or 1).

3.3.12 Final Combined Fusion Model (P_{final})

The final combined output integrates the base prediction, confidence scaling, and dynamic penalty terms:

$$P_{\text{final}} = [S \cdot (1 + \gamma \cdot cf)] / [1 + \alpha \cdot D + \beta \cdot U + \varepsilon] \quad (11)$$

Here, γ acts as the gain multiplier for confident predictions, α regulates the disagreement penalty, and β regulates the uncertainty penalty. The denominator term $(1 + \alpha D + \beta U)$ acts as a dynamic dampener that suppresses the output probability whenever model predictions conflict or exhibit high ambiguity.

3.3.13 End-to-End Workflow and Optimization

Step 1: Data Alignment: Ensure clinical probabilities, image probabilities, and ground-truth labels are matched row-by-row across samples.

Step 2: Probability Calibration: Convert raw inputs to log-odds space and re-project through the logistic sigmoid function to standardize the probability scale.

Step 3: Feature Engineering: Compute sample-level metric arrays for S , cf , D , and U .

Step 4: Hyperparameter Optimization: Perform grid-search cross-validation across hyperparameter ranges for α , β , and γ to maximize Balanced Accuracy.

Step 5: Decision Thresholding: Evaluate binary classification cutoffs (τ) on the fused probabilities to establish the optimal operating threshold.

Step 6: Model Evaluation: Generate validation statistics including Precision, Recall, F1-Score, ROC-AUC, and the Confusion Matrix.

3.3.14 Complete Technical Pseudocode Implementation

The reference implementation below computes the fusion metrics, performs grid-search optimization over the penalty parameters, and returns the final fused predictions with evaluation statistics.

```
# Multi-Modal Probability Fusion Implementation Framework
```

```
import numpy as np

from sklearn.metrics import balanced_accuracy_score, \
    precision_recall_fscore_support, roc_auc_score, confusion_matrix

def logit(p, eps=1e-7):
    """Map probabilities to log-odds space with numerical clipping."""
    p_clipped = np.clip(p, eps, 1.0 - eps)
    return np.log(p_clipped / (1.0 - p_clipped))

def sigmoid(z):
    """Standard logistic function."""
    return 1.0 / (1.0 + np.exp(-z))
```

```

def binary_entropy(p, eps=1e-7):
    """Compute Shannon binary cross-entropy."""
    p_clipped = np.clip(p, eps, 1.0 - eps)
    return -(p_clipped * np.log2(p_clipped)
            + (1.0 - p_clipped) * np.log2(1.0 - p_clipped))

def compute_fusion_metrics(Pc_prime, Pi_prime, eps=1e-7):
    """Derive the core mathematical fusion metrics: S, cf, D, U."""
    S = 0.5 * Pc_prime + 0.5 * Pi_prime
    cf = (Pc_prime**2 + Pi_prime**2) / (Pc_prime + Pi_prime + eps)
    D = np.abs(Pc_prime - Pi_prime)
    U = (binary_entropy(Pc_prime, eps) + binary_entropy(Pi_prime, eps)) / 2.0
    return S, cf, D, U

def execute_probability_fusion(Pc_raw, Pi_raw, labels,
                              alpha_grid, beta_grid, gamma_grid,
                              threshold_grid, eps=1e-7):
    """Optimize fusion parameters and return prediction metrics."""

    # Data alignment
    n = min(len(Pc_raw), len(Pi_raw), len(labels))
    Pc, Pi, y = np.array(Pc_raw[:n]), np.array(Pi_raw[:n]), np.array(labels[:n])

    # Calibration
    Pc_prime = sigmoid(logit(Pc, eps))
    Pi_prime = sigmoid(logit(Pi, eps))

    # Feature derivation
    S, cf, D, U = compute_fusion_metrics(Pc_prime, Pi_prime, eps)

    best_bacc = -1.0
    best_params = {}

    # Grid-search optimization

```



```

for alpha in alpha_grid:
    for beta in beta_grid:
        for gamma in gamma_grid:
            P_fused = (S * (1.0 + gamma * cf)) / (1.0 + alpha * D + beta * U + eps)

            for th in threshold_grid:
                preds = (P_fused >= th).astype(int)
                score = balanced_accuracy_score(y, preds)

                if score > best_bacc:
                    best_bacc = score
                    best_params = {'alpha': alpha, 'beta': beta,
                                   'gamma': gamma, 'threshold': th}

# Compute final predictions
a, b, g, th = (best_params['alpha'], best_params['beta'],
               best_params['gamma'], best_params['threshold'])
final_prob = (S * (1.0 + g * cf)) / (1.0 + a * D + b * U + eps)
final_pred = (final_prob >= th).astype(int)

# Metrics evaluation
precision, recall, f1, _ = precision_recall_fscore_support(
    y, final_pred, average='binary', zero_division=0)
auc = roc_auc_score(y, final_prob)
cm = confusion_matrix(y, final_pred)

return {
    'best_parameters': best_params,
    'final_probabilities': final_prob,
    'final_predictions': final_pred,
    'evaluation': {
        'balanced_accuracy': best_bacc,
        'precision': precision,
        'recall': recall,
        'f1_score': f1,
    }
}

```

```

        'roc_auc': auc,
        'confusion_matrix': cm,
    },
}

```

3.4 Experimental Setup

The experiments were conducted using the Kaggle Notebook platform with Python as the primary programming language. The implementation was performed using Scikit-learn for machine learning and TensorFlow/Keras for deep learning, while NumPy, Pandas, OpenCV, Matplotlib, and Seaborn were utilized for data processing, image processing, visualization, and performance analysis.

For the clinical prediction model, the dataset was divided into training and testing subsets using a stratified train-test split to preserve the original class distribution. To address class imbalance in the training data, the SMOTETomek sampling technique was applied before model training. The selected CatBoost model was then trained using the processed clinical features.

For the retinal fundus image prediction model, all images were resized to 224×224 pixels and normalized before training. Data augmentation techniques, including rotation, zoom, and horizontal flipping, were applied to improve model generalization. The selected DenseNet121 model was trained using the preprocessed retinal fundus images.

The prediction probabilities generated by the CatBoost and DenseNet121 models were subsequently integrated using the proposed Confidence, Agreement, and Uncertainty-aware Fusion (CAU-Fusion) framework. The performance of the proposed framework was evaluated using standard classification metrics, including accuracy, precision, recall, F1-score, ROC-AUC, balanced accuracy, false positive rate (FPR), and false negative rate (FNR).

3.5 Cost Analysis

The cost analysis of the proposed CAU-Fusion framework considers all resources utilized throughout the project lifecycle, including hardware, cloud computing, software, dataset, internet, electricity, cloud storage, and application deployment. Table 4 summarizes the estimated cost associated with each component.

Table 4: Estimated Cost Breakdown of the CAU-Fusion Framework

No.	Cost Item	Description	Estimated Cost (BDT)
1	Hardware (Depreciation)	Estimated proportional depreciation cost of the existing personal laptop (Intel i5-10300H, 8 GB RAM, NVIDIA GTX 1650) for the project duration (~3 months usage out of ~3-year lifespan)	1,500
2	Cloud Computing	Kaggle Notebook free-tier GPU (NVIDIA Tesla T4) used for model training	0 (Free Tier)
3	Software & Libraries	Python, Scikit-learn, TensorFlow, Keras, Pandas, NumPy, OpenCV, Matplotlib, Seaborn — all open-source and license-free	0 (Open-Source)

No.	Cost Item	Description	Estimated Cost (BDT)
4	Dataset Acquisition	Publicly available clinical records and retinal fundus image dataset obtained from open repositories	0 (Free/Public)
5	Cloud Storage & Backup	Google Drive / cloud storage used for backing up datasets, trained models, and notebooks	500
6	Internet Usage	Broadband internet connectivity for approximately 3 months (dataset access, Kaggle usage, research, deployment, documentation)	3,000
7	Electricity	Estimated electricity consumption for local system usage during development, training, and experimentation	1,200
8	Application Deployment	Streamlit Community Cloud / Hugging Face Spaces free-tier hosting for prototype demonstration; estimated cost reserved for a paid compute tier (required for stable performance with TensorFlow + CatBoost) and potential custom domain for professional presentation	1,500
Total Estimated Cost			7,700 BDT

The cost analysis shows that the majority of the resource requirements - cloud computing, software, and dataset - were fulfilled at no direct monetary cost through the use of free-tier cloud services and open-source tools. The primary quantifiable expenses were hardware depreciation, cloud backup, internet connectivity, electricity, and application deployment, bringing the total estimated project cost to approximately 7,700 BDT. The deployment cost specifically accounts for the additional compute resources typically required to reliably host a multi-model (CatBoost + DenseNet121) application beyond the limits of a free hosting tier. This demonstrates that the CAU-Fusion framework can be developed and evaluated as a research prototype with minimal financial investment, while also providing a realistic estimate of the resources needed to move toward a stable, publicly accessible deployment. For future large-scale healthcare deployment, additional costs would be required for secure cloud infrastructure, dedicated data storage, cybersecurity measures, regulatory compliance, and system maintenance; however, these fall outside the scope of the current research-stage cost analysis.

3.6 Summary

This chapter presented the materials, methodology, and experimental procedures adopted for the development of the proposed Confidence, Agreement, and Uncertainty-aware Fusion (CAU-Fusion) framework. It described the collection of paired clinical records and retinal fundus images and established their one-to-one correspondence using Image_ID. The chapter also discussed dataset exploration, including the analysis of clinical features and retinal image characteristics, followed by appropriate sampling and preprocessing techniques for both modalities. The research environment, computational resources, programming tools, and supporting libraries used for implementation were also presented. Furthermore, the chapter described the overall methodology of the CAU-Fusion framework, where the clinical and retinal image data were processed through separate pipelines. The selected CatBoost and DenseNet121 models were introduced for generating clinical and image prediction probabilities, respectively. The design of the proposed CAU-Fusion framework was then presented, including probability calibration, simple fusion, agreement, confidence, disagreement, entropy-based uncertainty, and the final combined fusion model. The mathematical formulation, end-to-end workflow, optimization process, and experimental setup were also discussed. These components provide a complete methodological foundation for the experimental evaluation and comparative analysis presented in the following chapter.

4.1 Obtained Result

The experimental results were obtained by training and evaluating multiple machine learning models on the clinical dataset and multiple deep learning models on the retinal fundus image dataset. For the clinical data, Logistic Regression, Random Forest, CatBoost, XGBoost, and MLP were evaluated. Among them, CatBoost achieved the best overall performance with 99.00% accuracy, 100.00% recall, and 99.41% F1-score. For the retinal image data, DenseNet121, EfficientNet-B0, InceptionV3, ResNet-50, and Xception were evaluated. Although InceptionV3 achieved the highest numerical accuracy of 99.67%, DenseNet121 was selected because of its balanced performance, high recall of 99.81%, AUC-ROC of 99.96%, stable convergence, and strong generalization. Therefore, CatBoost and DenseNet121 were selected as the final clinical and image models for the proposed framework.

4.1.1 Logistic Regression

The Logistic Regression model was implemented as the baseline classifier for predicting diabetic cases using the structured clinical dataset. Before training, the dataset was preprocessed by removing duplicate records, handling missing values, encoding categorical attributes, and balancing the class distribution using the SMOTETomek resampling technique. After preprocessing, the balanced dataset was divided into training and testing subsets. The experimental results indicate that Logistic Regression achieved an overall accuracy of 97.00%, with a precision of 97.65%, recall of 98.81%, and an F1-score of 98.22%. The model also achieved an AUC-ROC of 98.64%, demonstrating strong discriminative capability between diabetic and non-diabetic patients. Figure 8 and 9 shows that Logistic Regression produced reliable classification performance, its accuracy was comparatively lower than the ensemble learning models evaluated in this research. Nevertheless, it provided a strong baseline and demonstrated that the clinical features used in this study possess significant predictive capability for diabetes classification.

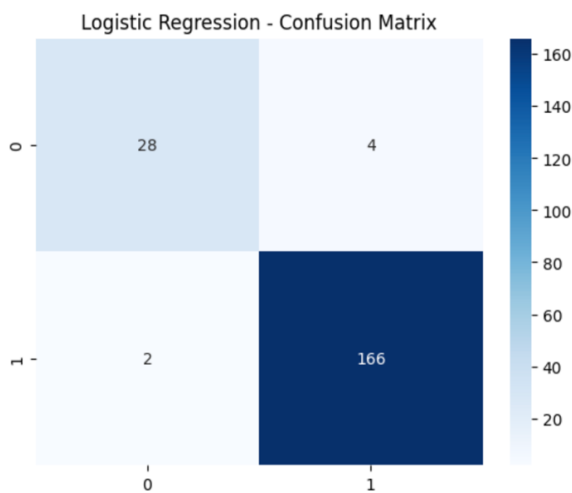


Figure 8: Confusion Matrix of Logistic Regression

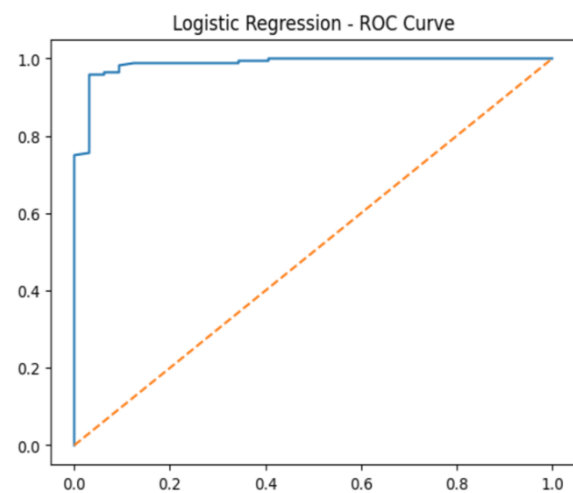


Figure 9: ROC Curve of Logistic Regression

4.1.2 Random Forest

The Random Forest model was evaluated to investigate the effectiveness of ensemble decision tree learning for structured clinical data classification. Multiple decision trees were trained using bootstrap sampling, and the final prediction was determined through majority voting, improving classification robustness and reducing overfitting. The model achieved an overall accuracy of 98.00%, together with a precision of 98.24%, recall of 99.40%, and an F1-score of 98.82%. The model also obtained an AUC-ROC of 97.28%, indicating excellent classification capability. Figure 10 and 11 shows that compared with Logistic Regression, Random Forest demonstrated improved predictive performance due to its ability to model nonlinear relationships among clinical variables. The results indicate that ensemble tree-based methods are highly effective for diabetes prediction using structured healthcare data.

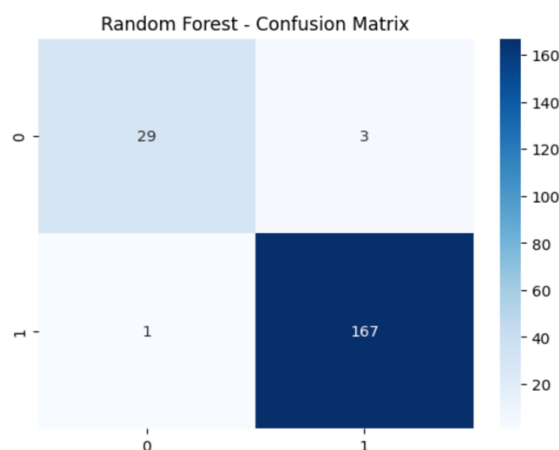


Figure 10: Confusion Matrix of Random Forest

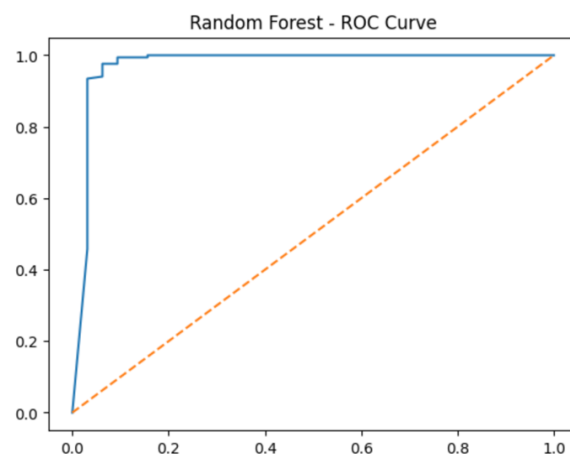


Figure 11: ROC Curve of Random Forest

4.1.3 CatBoost

CatBoost was implemented as the primary gradient boosting algorithm for structured clinical data classification. The algorithm efficiently handles complex feature interactions and categorical information while minimizing prediction bias through ordered boosting techniques. Following the preprocessing stage, the balanced clinical dataset was used to train and evaluate the CatBoost classifier. The experimental results demonstrate that CatBoost achieved the best overall performance among all machine learning algorithms evaluated in this study. The model obtained an overall accuracy of 99.00%, a precision of 98.82%, a perfect recall of 100.00%, and an F1-score of 99.41%. Furthermore, the model achieved an AUC-ROC of 98.72%, indicating excellent discrimination between diabetic and non-diabetic patients. One of the most significant findings is the model's 100% recall, indicating that every diabetic patient in the testing dataset was correctly identified without producing false negative predictions. Such performance is highly desirable for clinical decision support systems because missing diabetic patients may delay diagnosis and treatment. Figure 12 and 13 shows that XGBoost produced similar numerical performance, CatBoost demonstrated greater stability during training and handled the clinical dataset more effectively. Therefore, CatBoost was selected as the final machine learning model for the proposed hybrid diabetic retinopathy detection framework.

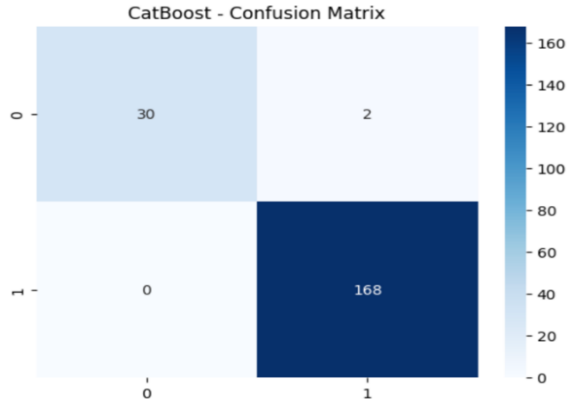


Figure 12: Confusion Matrix of CatBoost

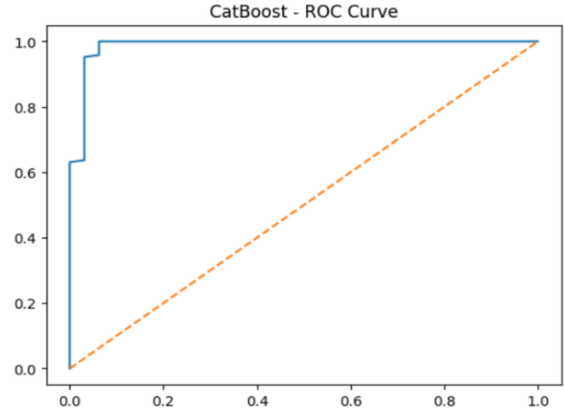


Figure 13: ROC Curve of CatBoost

4.1.4 XGBoost

The XGBoost model was implemented to evaluate the effectiveness of extreme gradient boosting for structured clinical data classification. The model utilizes sequential tree construction and gradient optimization to improve predictive accuracy while reducing classification errors. The model achieved an overall accuracy of 99.00%, together with a precision of 98.82%, a recall of 100.00%, and an F1-score of 99.41%. The corresponding AUC-ROC of 98.49% demonstrates excellent classification capability. Figure 14 and 15 shows that the results indicate that XGBoost performed almost identically to CatBoost in terms of classification accuracy and recall. However, CatBoost exhibited slightly better training stability and was therefore selected as the final machine learning model for the proposed framework.

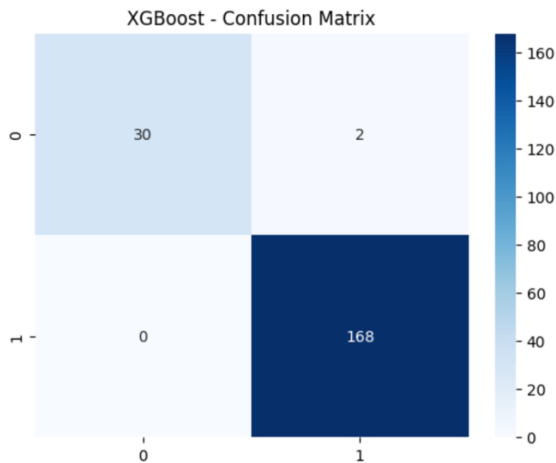


Figure 14: Confusion Matrix of XGBoost

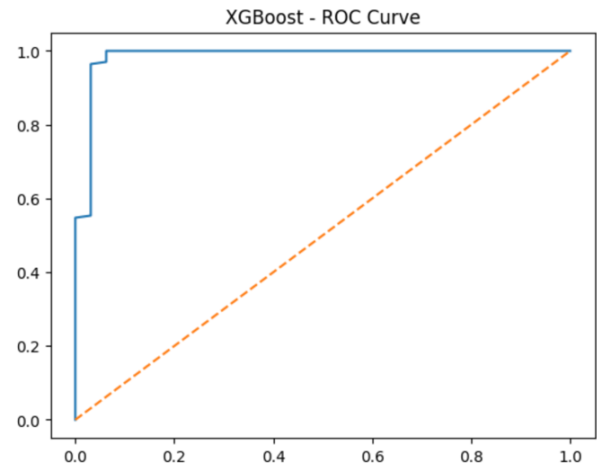


Figure 15: ROC Curve of XGBoost

4.1.5 Multi-Layer Perceptron (MLP)

The Multi-Layer Perceptron (MLP) model was implemented as a neural network-based classifier for predicting diabetic cases using structured clinical information. The network learned nonlinear relationships among the clinical variables through multiple hidden layers and backpropagation optimization. The experimental evaluation shows that the MLP model achieved an overall accuracy of 98.50%, with a precision of 98.25%, recall of 100.00%, and an F1-score of 99.12%. The model also achieved an AUC-ROC of 98.83%, indicating excellent classification performance. Figure 16 and 17 shows that MLP produced highly competitive results, CatBoost achieved slightly better overall performance and demonstrated superior stability during training. Therefore, CatBoost remained the preferred machine learning model for structured clinical data classification.

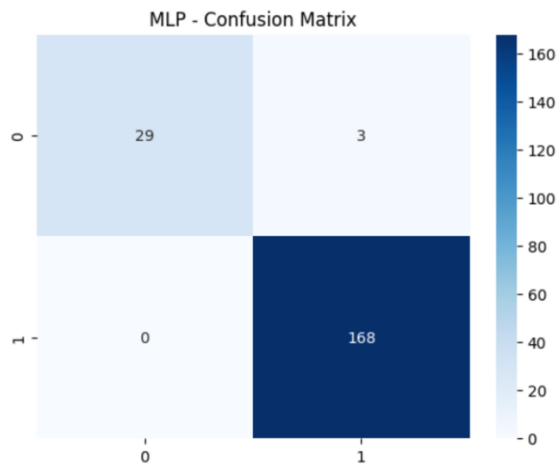


Figure 16: Confusion Matrix of MLP

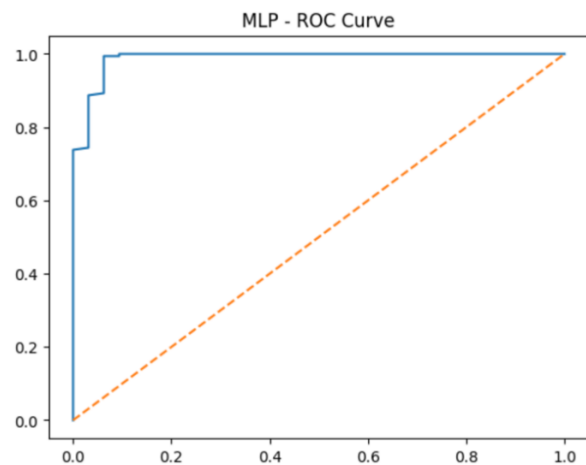


Figure 17 : ROC Curve of the MLP

4.1.6 DenseNet121

The DenseNet121 architecture was implemented using transfer learning to classify diabetic retinopathy from retinal fundus images. Before training, all retinal images were resized, normalized, and augmented using random transformations to improve the diversity of the training data and reduce overfitting. The dataset was divided into training and testing subsets using an 80:20 split. The pre-trained DenseNet-121 model was then fine-tuned to learn disease-specific retinal features while preserving the rich feature representations learned from large-scale image datasets. During training, the model exhibited stable convergence, with both training and validation accuracy increasing steadily across successive epochs. At the same time, the corresponding loss curves gradually decreased without significant fluctuations, indicating that the model learned meaningful retinal features while maintaining good generalization capability. The close agreement between the training and validation performance suggests that the model did not suffer from severe overfitting. The final experimental evaluation demonstrates that DenseNet-121 achieved an overall accuracy of 98.77%, together with a precision of 97.45%, recall of 99.81%, and an F1-score of 98.62%. The model also obtained a Balanced Accuracy of 98.89%, Specificity of 97.97%, and an AUC-ROC of 99.96%, indicating outstanding classification capability. The False Negative Rate of only 0.19% demonstrates that almost all diabetic cases were successfully detected. Figure 18 and 19 shows that Although InceptionV3 achieved slightly higher overall accuracy, DenseNet121 demonstrated more balanced diagnostic performance, stable convergence, excellent recall, and superior generalization throughout the training process. Therefore, DenseNet121 was selected as the final deep learning model for the proposed diabetic retinopathy detection framework.

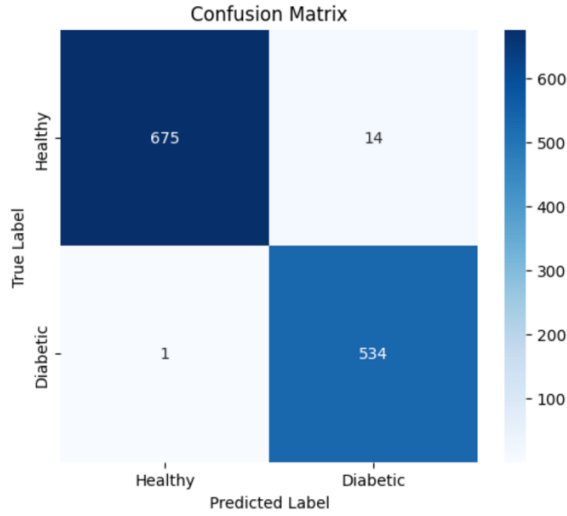


Figure 18: Confusion Matrix of DenseNet121

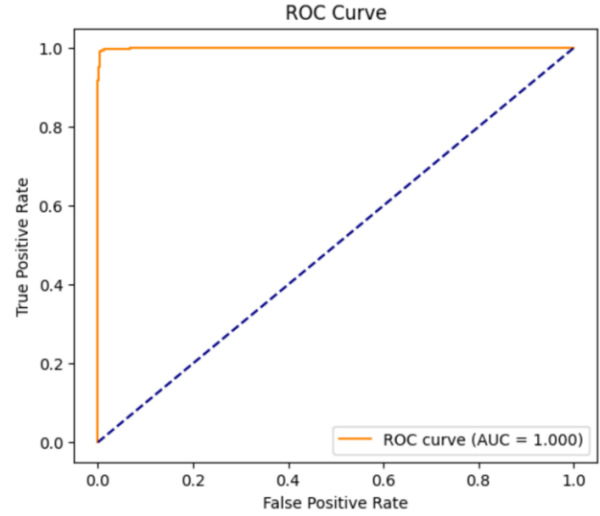


Figure 19: ROC Curve of DenseNet121

4.1.7 EfficientNet-B0

The EfficientNet-B0 model was implemented to evaluate the effectiveness of compound-scaled convolutional neural networks for retinal image classification. The retinal images were preprocessed using resizing, normalization, and augmentation before training. The dataset was divided using a 70:30 training and testing ratio. During training, the model showed only limited improvement in validation performance. Although the training process remained stable, the validation accuracy did not improve significantly, suggesting that the model failed to learn highly discriminative retinal features under the selected experimental configuration. The final testing results indicate that EfficientNet-B0 achieved an overall accuracy of 56.26%. However, the model produced 0.00% precision, 0.00% recall, and 0.00% F1-score, indicating that it failed to correctly identify diabetic retinal images. The model predicted nearly all samples as belonging to the healthy class, resulting in a Specificity of 100% but a False Negative Rate of 100%. The corresponding AUC-ROC of 57.83% further confirms the poor discriminative capability of the model. Figure 20 and 21 shows that the experimental findings indicate that EfficientNet-B0 was unable to converge to an effective solution under the current training configuration. Therefore, this architecture was not considered suitable for the final proposed framework.

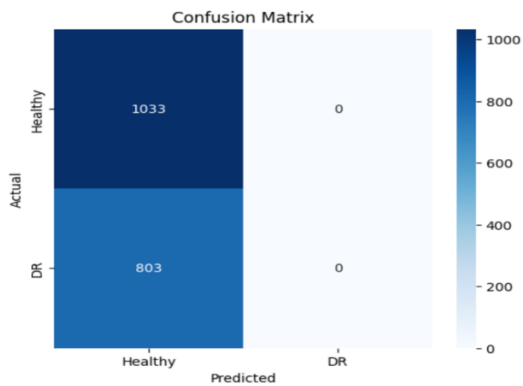


Figure 20: Confusion Matrix of EfficientNet-B0

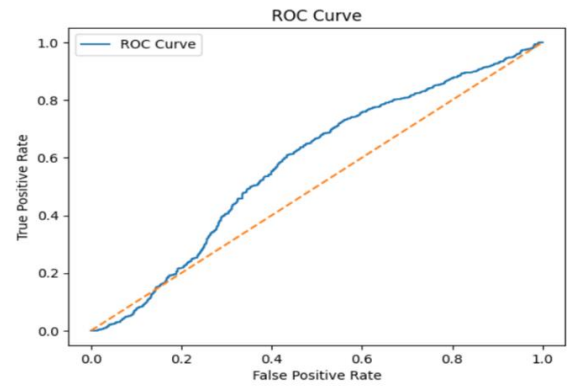


Figure 21: ROC Curve of EfficientNet-B0

4.1.8 InceptionV3

The InceptionV3 model was implemented using transfer learning to classify diabetic retinopathy from retinal fundus images. Retinal images were resized, normalized, and augmented before training, while an 80:20 train-test split was used throughout the experiment. The pre-trained InceptionV3 architecture was fine-tuned to learn retinal disease characteristics from the image dataset. The model demonstrated rapid convergence during training. Both the training and validation accuracy increased consistently, while the corresponding loss values gradually decreased. The learning curves indicate that the model successfully learned highly discriminative retinal representations with minimal overfitting. The final evaluation results indicate that InceptionV3 achieved the highest numerical accuracy of 99.67%, together with a precision of 99.44%, recall of 99.81%, F1-score of 99.63%, and an AUC-ROC of 99.998%. These results demonstrate exceptional classification capability and excellent separation between diabetic and healthy retinal images. Figure 22 and 23 shows that InceptionV3 achieved the highest overall accuracy among all evaluated CNN architectures, the proposed framework selected DenseNet-121 as the final deep learning model because of its more balanced performance, stable convergence, and consistent generalization across the entire training process.

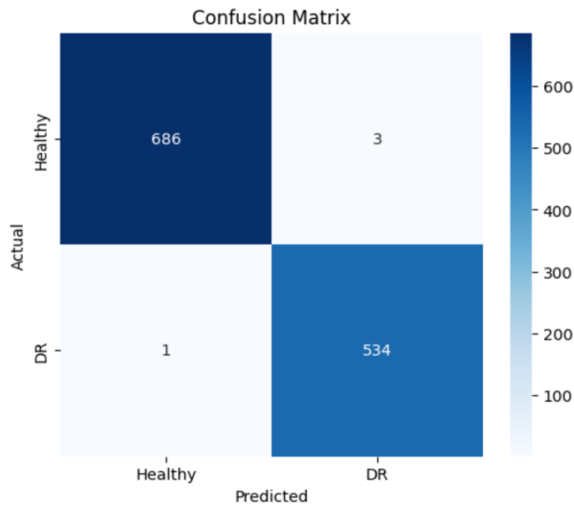


Figure 22: Confusion Matrix of InceptionV3

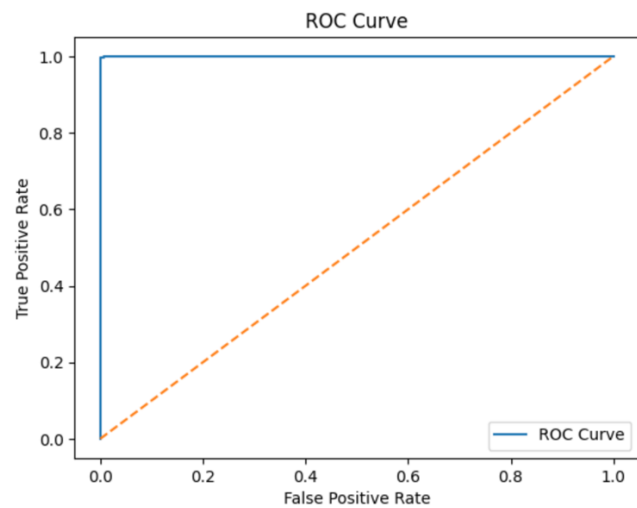


Figure 23: ROC Curve of InceptionV3

4.1.9 ResNet-50

The ResNet-50 architecture was evaluated to investigate the effectiveness of deep residual learning for diabetic retinopathy classification. The retinal images were preprocessed through resizing, normalization, and augmentation before model training. During training, the model exhibited gradual convergence with steady improvements in both training and validation accuracy. However, compared with DenseNet-121 and InceptionV3, the validation performance remained comparatively lower, indicating relatively weaker generalization. The final testing results demonstrate that ResNet-50 achieved an overall accuracy of 87.17%, with a precision of 90.73%, recall of 78.69%, and an F1-score of 84.28%. The model also achieved an AUC-ROC of 90.57% and a Specificity of 93.76%. However, the relatively high False Negative Rate of 21.31% indicates that a considerable number of diabetic retinal images were incorrectly classified as healthy. Figure 24 and 25 shows that ResNet-50 produced satisfactory classification performance, its lower recall and higher false negative rate make it less suitable for medical diagnosis compared with DenseNet121.

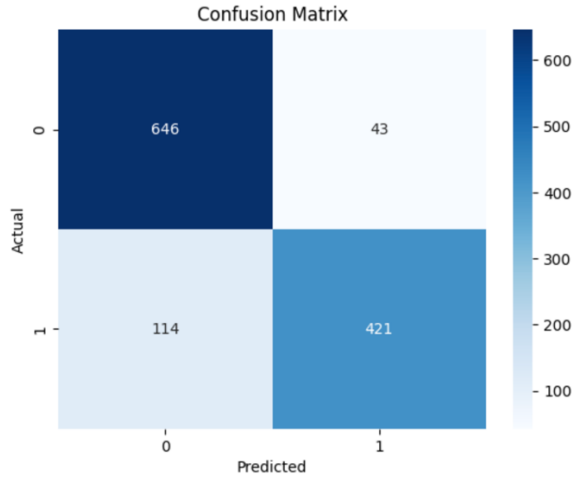


Figure 24: Confusion Matrix of ResNet-50

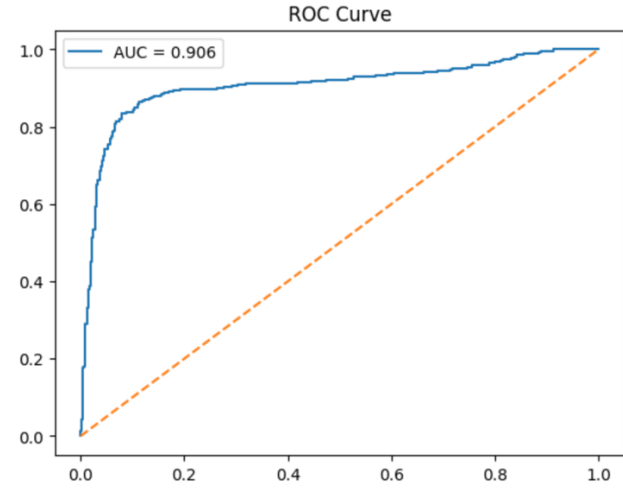


Figure 25: ROC Curve of ResNet-50

4.1.10 Xception

The Xception architecture was implemented using transfer learning to evaluate the effectiveness of depth wise separable convolutions for retinal fundus image classification. Standard preprocessing techniques, including resizing, normalization, and data augmentation, were applied before model training. During the training process, the model demonstrated stable convergence with consistent improvements in both training and validation accuracy. The corresponding loss curves gradually decreased throughout training, indicating effective learning without significant overfitting. The final experimental evaluation shows that Xception achieved an overall accuracy of 98.37%, together with a precision of 97.07%, recall of 99.25%, F1-score of 98.15%, and an AUC-ROC of 99.92%. The model also achieved a Balanced Accuracy of 98.47% while maintaining a very low False Negative Rate of only 0.75%. Figure 26 and 27 shows that the experimental findings confirm that Xception is a highly effective deep learning architecture for retinal disease classification. Nevertheless, considering the overall balance of accuracy, recall, generalization capability, and training stability, DenseNet-121 was selected as the final image classification model for the proposed framework.

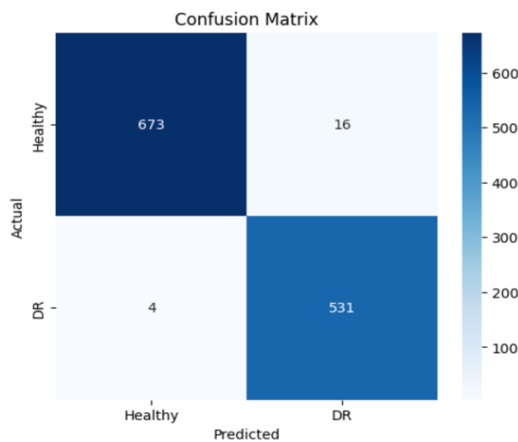


Figure 26: Confusion Matrix of Xception

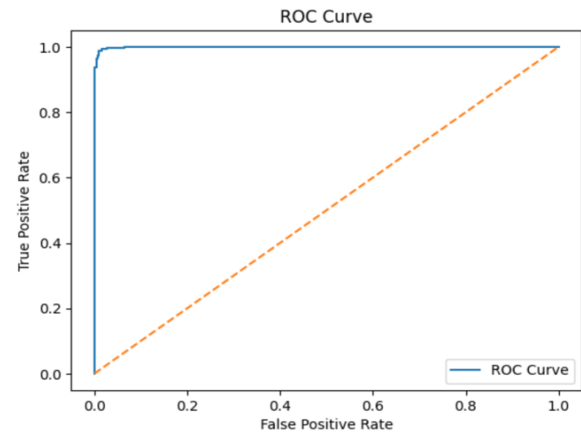


Figure 27: ROC Curve of the Xception

Table 2 presents the comparative performance of the five machine learning models developed using the structured clinical dataset. The models were evaluated based on Accuracy, Precision, Recall, F1-score, and AUC-ROC to determine the most suitable algorithm for diabetes prediction from clinical information. Among the evaluated models, CatBoost achieved the best overall performance with an accuracy of 99.00%, precision of 98.82%, perfect recall of 100.00%, F1-score of 99.41%, and an AUC-ROC of 98.72%. These results indicate that CatBoost successfully identified all diabetic patients while maintaining excellent overall classification performance. Although XGBoost achieved similar numerical performance, CatBoost demonstrated greater training stability and more effective handling of structured clinical features. Random Forest and MLP also produced excellent classification results, while Logistic Regression served as a strong baseline model with an accuracy of 97.00%. Based on the overall evaluation, CatBoost was selected as the final machine learning model for the proposed framework due to its superior predictive capability and robustness.

Table 2: Performance Comparison of Using the Clinical Dataset

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC-ROC (%)
Logistic Regression	97.00	97.65	98.81	98.22	98.64
Random Forest	98.00	98.24	99.40	98.82	97.28
CatBoost	99.00	98.82	100.00	99.41	98.72
XGBoost	99.00	98.82	100.00	99.41	98.49
MLP	98.50	98.25	100.00	99.12	98.83

Table 3 presents the comparative performance of the five transfer learning models implemented for diabetic retinopathy classification using retinal fundus images. The models were assessed using the same evaluation metrics to ensure a fair comparison of their diagnostic capabilities. Among the evaluated architectures, InceptionV3 achieved the highest numerical performance with an accuracy of 99.67%, F1-score of 99.63%, and an AUC-ROC of 99.998%. However, DenseNet-121 demonstrated a more balanced diagnostic performance by achieving an accuracy of 98.77%, precision of 97.45%, recall of 99.81%, F1-score of 98.62%, and an AUC-ROC of 99.96%, while maintaining stable convergence and excellent generalization throughout the training process. Xception also achieved outstanding classification performance, whereas ResNet-50 produced comparatively lower recall, indicating that more diabetic cases were missed during prediction. EfficientNet-B0 recorded the weakest performance because it failed to effectively distinguish diabetic cases under the selected experimental configuration. Although InceptionV3 achieved slightly higher numerical metrics, DenseNet-121 was selected as the final deep learning model for the proposed framework because it provided a better balance between accuracy, sensitivity, stability, and generalization capability. Its consistently high recall and very low false negative rate make it more suitable for practical diabetic retinopathy screening, where correctly identifying disease cases is more critical than achieving marginal improvements in overall accuracy.

Table 3: Performance Comparison of Using the Image Dataset

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC-ROC (%)
DenseNet-121	98.77	97.45	99.81	98.62	99.96
EfficientNet-B0	56.26	0.00	0.00	0.00	57.83
InceptionV3	99.67	99.44	99.81	99.63	99.998
ResNet-50	87.17	90.73	78.69	84.28	90.57
Xception	98.37	97.07	99.25	98.15	99.92

4.2 In-depth Result Analysis

The experimental results demonstrate that both the machine learning models trained on structured clinical data and the deep learning models trained on retinal fundus images achieved promising performance for diabetic retinopathy prediction. However, considerable differences were observed among the evaluated algorithms in terms of classification accuracy, sensitivity, specificity, and overall diagnostic capability. For the structured clinical dataset, five machine learning algorithms were evaluated, including Logistic Regression, Random Forest, CatBoost, XGBoost, and Multi-Layer Perceptron (MLP). Among these models, CatBoost achieved the best overall performance with an accuracy of 99.00%, perfect recall (100%), and an F1-score of 99.41%. The model also maintained excellent precision and AUC-ROC values, demonstrating its ability to correctly identify diabetic patients while minimizing false negative predictions. Since early diagnosis is the primary objective of diabetic screening systems, CatBoost was selected as the final machine learning model for structured clinical data.

For retinal image classification, five transfer learning architectures were investigated, namely DenseNet-121, EfficientNet-B0, InceptionV3, ResNet-50, and Xception. Although InceptionV3 achieved the highest numerical accuracy (99.67%), the performance difference compared with DenseNet-121 was relatively small. DenseNet-121 demonstrated an excellent balance between classification accuracy (98.77%), recall (99.81%), F1-score (98.62%), and AUC-ROC (99.96%), while maintaining stable convergence throughout the training process. Its confusion matrix also showed consistent classification of both diabetic and healthy retinal images with very few misclassifications. Xception also achieved excellent performance with an accuracy of 98.37% and an AUC-ROC of 99.92%, confirming its capability for retinal disease classification. ResNet-50 produced acceptable results but showed comparatively lower recall, indicating that more diabetic cases were missed than in DenseNet-121. In contrast, EfficientNet-B0 performed poorly under the selected experimental settings, predicting almost all samples as belonging to a single class, resulting in very low recall and F1-score.

Considering the overall experimental findings, DenseNet-121 was selected as the final deep learning architecture for the proposed framework. Although InceptionV3 achieved marginally higher accuracy, DenseNet-121 demonstrated a more balanced and reliable diagnostic performance with excellent sensitivity, high AUC-ROC, stable convergence, and strong generalization capability. These characteristics make DenseNet-121 more appropriate for real-world diabetic retinopathy screening, where minimizing missed diagnoses is more important than achieving a slightly higher overall accuracy.

4.3 Performance Evaluation

The proposed framework was evaluated using multiple performance metrics, including Accuracy, Precision, Recall, F1-score, Balanced Accuracy, Specificity, False Positive Rate (FPR), False Negative Rate (FNR), and Area Under the Receiver Operating Characteristic Curve (AUC-ROC). These metrics provide a comprehensive assessment of diagnostic performance for both structured clinical data and retinal fundus image classification. For the structured clinical dataset, CatBoost achieved the best overall performance among all machine learning algorithms. The model obtained an overall accuracy of 99.00%, a recall of 100%, and an F1-score of 99.41%, indicating its excellent capability for identifying diabetic patients without missing positive cases. The strong AUC-ROC further demonstrates the robustness of CatBoost for structured healthcare data.

Among the image-based deep learning models, DenseNet-121 consistently demonstrated highly reliable performance across all evaluation metrics. The model achieved an overall accuracy of 98.77%, a recall of 99.81%, a Balanced Accuracy of 98.89%, and an AUC-ROC of 99.96%. These results indicate that DenseNet-121 successfully detected almost all diabetic retinal images while maintaining excellent classification performance for healthy cases. Although InceptionV3 produced a slightly higher classification

accuracy, the difference was relatively small. DenseNet-121 maintained a more balanced diagnostic performance while exhibiting stable learning behaviour throughout the training process. Such characteristics are particularly desirable for medical diagnosis systems where robustness, consistency, and sensitivity are more important than marginal improvements in accuracy alone. Xception also demonstrated excellent classification capability with high accuracy and AUC values, whereas ResNet-50 achieved moderate performance with comparatively lower sensitivity. EfficientNet-B0 produced unsatisfactory results and therefore was excluded from the final proposed framework. Based on the overall evaluation, CatBoost was selected as the final machine learning model for structured clinical data, while DenseNet121 was selected as the final deep learning architecture for retinal fundus image classification.

4.4 Discussion

The experimental results confirm that both structured clinical information and retinal fundus images contain highly informative features for automated diabetic retinopathy detection. Machine learning algorithms proved highly effective for structured patient data, whereas transfer learning architectures demonstrated remarkable capability for retinal image analysis. Among the machine learning models, CatBoost consistently outperformed Logistic Regression, Random Forest, XGBoost, and MLP. Its perfect recall indicates that all diabetic patients were correctly identified during testing, making it highly suitable for clinical decision support systems where minimizing false negative diagnoses is critical. For retinal image classification, DenseNet-121 demonstrated the most balanced and reliable performance among the evaluated convolutional neural networks. Although InceptionV3 achieved slightly higher numerical accuracy, DenseNet-121 consistently maintained excellent recall, strong AUC-ROC, stable convergence, and robust generalization across unseen retinal images. The dense connectivity mechanism of DenseNet-121 enables efficient feature reuse and improved gradient propagation, allowing the network to learn discriminative retinal features more effectively. Consequently, DenseNet121 was selected as the final image classification model for the proposed framework. The poor performance of EfficientNet-B0 indicates that not every state-of-the-art architecture performs equally well under identical experimental conditions. Future work may investigate improved hyperparameter tuning, extended training, or larger datasets to further enhance its performance. Similarly, additional optimization may improve the diagnostic sensitivity of ResNet-50.

Overall, the proposed hybrid framework successfully combines CatBoost for structured clinical information and DenseNet121 for retinal fundus image analysis, enabling accurate diabetic retinopathy prediction using two complementary sources of medical information. This multimodal approach has the potential to improve early diagnosis and support ophthalmologists in real-world clinical screening.

4.5 Summary

This chapter presented the experimental evaluation and comparative analysis of the proposed diabetic retinopathy detection framework. Five machine learning models were evaluated using clinical data, while five deep learning models were evaluated using retinal fundus images. Among the clinical models, CatBoost achieved the best performance with 99.00% accuracy, 100% recall, and a 99.41% F1-score. For image classification, although InceptionV3 achieved slightly higher accuracy, DenseNet121 was selected due to its balanced performance, high recall, excellent AUC-ROC, stable convergence, and strong generalization. Therefore, CatBoost and DenseNet121 were selected as the final clinical and image models for the proposed multimodal framework.

Chapter 5

Social and Environmental Impact of Engineering

The proposed Confidence, Agreement, and Uncertainty-aware Fusion (CAU-Fusion) framework is developed by integrating clinical tabular data and retinal fundus images to support reliable diabetes prediction. By combining information from two different data modalities, the framework aims to provide a more comprehensive approach than relying on a single source of information. The proposed system has potential impacts across several important domains, including societal, healthcare, technological, ethical, economic, and environmental aspects. The following sections discuss these potential impacts of the proposed engineering solution.

5.1 Societal Impact

Diabetes is a common chronic disease that can lead to serious health complications if it is not identified and managed properly at an early stage. Early risk identification can help individuals become more aware of their health condition and encourage them to take appropriate preventive measures. The proposed CAU-Fusion framework combines clinical information with retinal fundus images, providing a multimodal approach for diabetes prediction.

The framework has the potential to support early screening and health awareness, particularly in rural and underserved areas where access to specialized healthcare professionals and advanced diagnostic facilities may be limited. An automated decision-support approach could help identify individuals who may require further medical attention or additional clinical assessment.

Furthermore, early identification of potential health risks may encourage patients to maintain regular health monitoring and adopt preventive measures, such as lifestyle modification, improved dietary habits, physical activity, and regular medical consultation. Therefore, the proposed framework has the potential to contribute to greater health awareness and support improved access to preliminary screening technologies.

2. Healthcare Impact

The CAU-Fusion framework has potential applications as a decision-support tool for healthcare professionals. The framework processes two complementary types of medical information. CatBoost analyzes structured clinical features, while DenseNet121 analyzes retinal fundus images. The prediction probabilities generated by these two models are subsequently integrated through the CAU-Fusion framework.

Unlike simple probability combination, the proposed fusion strategy considers confidence, agreement, disagreement, and uncertainty during the fusion process. This approach can provide additional information regarding the reliability and consistency of predictions obtained from the two modalities. Such information may assist healthcare professionals when evaluating potentially high-risk patients.

The proposed system may also help improve the efficiency of preliminary screening by supporting faster analysis of available clinical and image-based information. However, the framework is not intended to replace doctors, medical specialists, or professional clinical diagnosis. Instead, it is designed to provide additional computational support for healthcare assessment. Final medical decisions should always remain under the supervision and judgment of qualified healthcare professionals.

5.3 Technological Impact

This project demonstrates the practical application of multimodal machine learning by integrating clinical tabular data and retinal fundus images within a unified prediction framework. Different data modalities have different characteristics and therefore require appropriate analytical approaches. In this project, CatBoost is used for structured clinical data, while DenseNet121 is used for retinal image analysis.

The proposed CAU-Fusion framework represents an additional technological contribution by integrating prediction probabilities through reliability-related components. The framework considers confidence, agreement, disagreement, and entropy-based uncertainty to refine the fusion process. This allows the system to consider not only the prediction values generated by individual models but also the relationship and reliability of those predictions.

The overall approach demonstrates how machine learning and deep learning models can be combined to process different types of healthcare data. The proposed methodology may potentially be adapted for other multimodal healthcare applications where structured medical records, medical images, or other complementary sources of information are available.

5.4 Ethical Impact

Ethical considerations are important when developing systems that process healthcare-related information. The proposed framework uses clinical features and retinal fundus images for prediction while avoiding the direct use of personal identity information within the model inputs. This approach supports the protection of patient privacy during the research and model development process.

However, privacy protection should remain an important consideration for any future real-world implementation. Medical data should be collected, stored, processed, and shared responsibly using appropriate security and privacy measures. Unauthorized access to sensitive healthcare information should be prevented through proper data management practices.

Another important ethical consideration is the role of artificial intelligence in medical decision-making. The proposed CAU-Fusion framework is designed as a decision-support system and should not be considered a replacement for professional medical judgment. AI-generated predictions may contain errors or uncertainty; therefore, qualified healthcare professionals should remain responsible for interpreting results and making final clinical decisions.

Future deployment should also consider issues such as fairness, transparency, data bias, and responsible use of AI in healthcare environments.

5.5 Economic Impact

The proposed framework may have potential economic benefits by supporting earlier identification of individuals who may be at risk of diabetes-related health conditions. Early screening and timely medical attention may help reduce the economic burden associated with delayed diagnosis and the development of severe complications.

The framework may also support more efficient healthcare screening by providing automated analysis of clinical and retinal image data. This could potentially reduce some repetitive manual workload and help healthcare professionals focus their attention on patients requiring further examination.

From a development perspective, the project is relatively cost-efficient because it utilizes Python and widely available open-source machine learning and deep learning libraries. Libraries and frameworks such as Scikit-learn, TensorFlow, Keras, Pandas, NumPy, and OpenCV reduce the need for expensive proprietary software.

In addition, the use of cloud-based computational resources, such as Kaggle Notebook, provides access to GPU resources without requiring the purchase of dedicated high-performance computing hardware. This makes research and prototype development more accessible and potentially beneficial for resource-limited environments.

5.6 Environmental Impact

The proposed CAU-Fusion framework supports the concept of digital healthcare analysis by enabling computational processing of clinical information and retinal fundus images. In potential future applications, digital and remote analysis could reduce dependence on some traditional physical diagnostic processes and unnecessary hospital visits.

Remote or digitally supported healthcare services may also reduce transportation requirements for patients who need preliminary screening or follow-up assessment. A reduction in unnecessary travel could contribute to lower fuel consumption and associated environmental impacts.

The use of cloud-based platforms such as Kaggle Notebook also provides access to shared computational infrastructure. Instead of requiring dedicated high-performance hardware for every stage of research and prototype development, computational resources can be accessed when required. This can support more efficient utilization of computing resources.

However, machine learning and deep learning models also require computational power and energy, particularly during model training. Therefore, efficient model development, appropriate use of computational resources, and avoiding unnecessary repeated training are important considerations for reducing the environmental impact of AI-based systems. Overall, the proposed framework demonstrates how digital healthcare technologies can potentially support more efficient healthcare processes while encouraging responsible and efficient use of computational resources.

6.1 Overall Contributions

This project presents CAU-Fusion, a reliability-aware multimodal framework for diabetic retinopathy prediction that integrates structured clinical data using CatBoost and retinal fundus images using DenseNet-121. Unlike conventional probability-based fusion, the framework incorporates prediction confidence, inter-modal disagreement, and uncertainty to improve the reliability of the final prediction. CatBoost achieved 99.00% accuracy with 100.00% recall, while DenseNet-121 achieved 98.77% accuracy, 99.81% recall, and 99.96% AUC-ROC, demonstrating strong unimodal performance. The framework further applies hyperparameter and decision-threshold optimization to dynamically handle conflicting or uncertain predictions. By combining complementary clinical and visual evidence while explicitly considering reliability factors, CAU-Fusion provides a more interpretable and trustworthy approach to multimodal diabetic retinopathy prediction.

6.2 Limitations and Future Works

Although CAU-Fusion provides a reliability-aware approach to multimodal diabetic retinopathy prediction, several limitations remain. The framework currently requires both structured clinical data and corresponding retinal fundus images, so missing or incomplete modalities may affect prediction performance. Moreover, the evaluation was conducted on the available dataset and computational environment, limiting generalizability to larger and more diverse clinical populations, while external clinical validation is still required. Future work can focus on larger multi-center datasets, external validation, missing-modality handling, and integration of additional modalities such as OCT, OCTA, laboratory results, and patient history. Further improvements can include advanced probability calibration and uncertainty estimation, enhanced explainability using clinical feature importance and retinal visual explanations, and real-time clinical decision support. The CAU-Fusion framework can also be evaluated on other multimodal medical diagnosis tasks to assess its broader applicability.

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Appendix

Mapping of Course and Program Outcomes

CSE400-A

PO1 (Engineering Knowledge): This project applies core knowledge of machine learning, deep learning, and probability theory to solve a real-world medical diagnosis problem. Concepts from statistics (probability calibration, Shannon entropy), gradient-boosted decision trees (CatBoost), and convolutional neural networks (DenseNet121) were integrated to design the CAU-Fusion framework, demonstrating the application of mathematical and engineering fundamentals to build a functioning diagnostic system.

PO4 (Investigation): The project involved systematic investigation into the limitations of standard fusion techniques through literature review, followed by controlled experimentation with different fusion formulations. Numeric worked examples and edge-case analysis (e.g., high-disagreement scenarios) were used to investigate and validate the behavior of the proposed formula before deployment.

CO	CO Description	PO	How PO was Addressed	Learning Domain
CO1	Integrate new and previously acquired knowledge for identifying a real-life complex computer science and engineering problem as the capstone project.	PO1: Engineering Knowledge	EP1: Defining the Engineering Problem: Applied theoretical knowledge of AI, deep learning, and multimodal data analysis to identify and design a real-world healthcare solution. Integrated patient clinical data and retinal fundus images to develop a unified feature-level multimodal fusion network for early diabetes prediction.	Cognitive
CO2	Examine various problem domains through literature review, define the problems, and formulate the objectives for the capstone project.	PO4: Investigation	EP1: Defining the Engineering Problem: Implemented state-of-the-art machine learning (ML), Logistic Regression, Random Forest, CatBoost and deep learning models (CNN – EfficientNet-B0) for analyzing clinical and image data. Performed Exploratory Data Analysis (EDA), applied feature-level fusion, and used explainable AI techniques (Grad-CAM, SHAP) to ensure interpretability, robustness, and clinically meaningful predictions.	Cognitive, Psychomotor

CSE400-B

PO2 (Problem Analysis): The problem of unreliable multi-modal medical diagnosis was analyzed by identifying a specific gap in existing systems — that naive probability averaging fails to account for model confidence and inter-modality disagreement. This was broken down into measurable sub-problems (confidence quantification, disagreement measurement, uncertainty estimation), each formulated mathematically before being combined into a complete fusion solution.

PO3 (Design/Development of Solutions): A complete solution was designed and implemented, consisting of two independently trained classifiers (a CatBoost model for clinical biomarkers and a DenseNet121 model for retinal fundus images) combined through a custom-designed, parameterized fusion equation (CAU-Fusion). The solution was further developed into a deployable, interactive web application with patient data entry, risk visualization, and automated PDF report generation.

PO5 (Modern Tool Usage): Modern machine learning and software engineering tools were used throughout, including CatBoost and TensorFlow/Keras for model development, Streamlit for interactive application deployment, ReportLab for automated PDF report generation, and cloud deployment platforms for making the system publicly accessible.

PO6 (The Engineer and Society): As a medical diagnostic support tool, this project directly addresses a societal healthcare need improving the reliability of diabetes and diabetic retinopathy screening, which is particularly valuable in resource-limited settings where access to specialist ophthalmologists may be restricted. The system is designed to assist, not replace, clinical judgment, reflecting responsible engineering practice in a healthcare context.

CO	CO Description	PO	PO Description	Learning Domains	Learning Subdomains	Rubric Design	Remarks
CO3	Analyze various aspects of the problem solution design	PO1	Engineering Knowledge	Cognitive	C2	Able to understand and define problem	Identified limitation of single data source (only CSV or only image) for diabetes prediction
					C3	Able to analyze problem components	Analyzed tabular data and image data separately and understood their contribution

					C4	Able to evaluate different approaches	Compared multiple ML models (LR, RF, XGBoost, CatBoost) and CNN models (ResNet, EfficientNet, Xception, DenseNet)
					C5	Able to make decisions	Selected CatBoost and DenseNet based on performance
CO4	Design and develop an effective solution	PO3	Design / Development of Solutions	Cognitive , Psychomotor	C6	Able to design complete system	Designed multimodal architecture combining CSV and image models
					P3	Able to implement solution practically	Implemented CatBoost model, DenseNet model, and fusion mechanism
					C5	Able to improve solution	Applied confidence-aware fusion instead of simple averaging

CO5	Apply modern tools for solving problem	PO5	Modern Tool Usage	Cognitive , Psychomotor	P2	Able to use tools properly	Used Python libraries (Scikit-learn, TensorFlow, OpenCV) for development
					P3	Able to use tools independently	Trained ML and DL models independently in Google Colab
					C3	Able to analyze using tools	Performed EDA, preprocessing, model evaluation, and visualization
CO6	Understand professional responsibilities in engineering practice	PO6	The Engineer and Society	Cognitive , Affective	C3	Able to relate engineering solution to real-world	Addressed early diabetes detection and healthcare improvement
					A3	Able to understand social responsibility	Considered ethical use of medical data and role of AI as support system

CSE400-C

PO7 (Environment and Sustainability): The system was optimized for lightweight deployment (e.g., replacing heavier model components with efficient alternatives to reduce computational and memory footprint), reflecting a sustainability-conscious approach to AI deployment that reduces energy and resource consumption compared to unnecessarily heavy model architectures.

PO8 (Ethics): The project embeds ethical safeguards directly into its design the system never outputs a falsely confident diagnosis, and instead explicitly flags conflicting or uncertain evidence for manual clinical review. Every report includes a clear disclaimer stating that the tool is AI-assisted and not a substitute for professional medical diagnosis, reflecting a commitment to responsible and transparent AI use in healthcare.

PO9 (Individual Work and Teamwork): The project required independent research, mathematical formulation, and implementation of the fusion framework, while also involving iterative feedback, debugging, and refinement cycles that reflect collaborative engineering practices typical of real-world software development environments.

PO10 (Communication): Findings were communicated through multiple formats a formal technical report with mathematically notated equations, a concept guide explaining every fusion term for defense preparation, an academic abstract, and a working demonstration application reflecting the ability to convey complex technical ideas to both technical and non-technical audiences.

PO11 (Project Management and Finance): The project was managed through defined phases literature review, model development, fusion formula design, application development, and deployment with iterative debugging and version control practices. Cost-effective, open-source tools were deliberately chosen throughout to keep the project financially accessible and reproducible.

PO12 (Life-Long Learning): The project required self-directed learning of new concepts beyond the standard curriculum, including probability calibration techniques, Shannon entropy in the context of model uncertainty, evidential/Dempster-Shafer fusion theory, and modern deployment practices reflecting a continued commitment to learning and adapting to new technical domains.

CO	Details	Knowledge Profile (K)	Engineering Problem (EP)
CO7	Societal and Environmental Contexts	K7: Demonstrate comprehension of engineering in society and its societal and environmental contexts. The proposed CAU-Fusion framework is developed to support early diabetic retinopathy prediction by integrating retinal fundus images and structured clinical data. Early detection enables timely clinical intervention, reducing the risk of irreversible vision loss and improving	EP2: Societal and Environmental Contexts. Conflicting Requirements: Diagnostic accuracy vs. computational complexity. Although the proposed framework aims to improve predictive reliability through confidence, agreement, and uncertainty estimation, integrating multiple predictive models increases computational requirements and deployment costs. Data accessibility vs. patient privacy: The system

		patients' quality of life. The framework contributes to public healthcare by assisting ophthalmologists in making more reliable diagnostic decisions, particularly in regions where specialist resources are limited. The adoption of trustworthy multimodal prediction systems can reduce diagnostic delays, improve screening efficiency, and support large-scale diabetic retinopathy screening programs.	requires access to sensitive retinal images and clinical records, which must be protected under healthcare data protection regulations. Healthcare accessibility: The framework should remain scalable and affordable for hospitals and diagnostic centers with limited computational resources. The system should also support ethical deployment by ensuring secure data management, fairness across patient populations, and compliance with applicable healthcare standards.
CO8	Ethical Principles and Professional Responsibilities	K7: Apply ethical principles and professional responsibilities in engineering practice. The proposed framework is intended to assist clinicians rather than replace professional medical judgment. Confidence, agreement, and uncertainty estimation are incorporated to improve transparency and help clinicians better understand the reliability of AI-generated predictions. The framework encourages responsible AI usage by promoting trustworthy decision support while reducing the risk of overconfident automated diagnosis.	EP6: Ethical Principles and Practices. Medical AI systems must ensure fairness, transparency, accountability, and patient safety. Retinal fundus images and structured clinical records contain sensitive patient information that requires secure storage, controlled access, and informed consent before use. The framework should minimize algorithmic bias across different demographic groups and avoid discrimination caused by imbalanced datasets. Prediction outputs should remain interpretable so clinicians can understand the reliability of the generated diagnosis before making clinical decisions. Compliance with healthcare regulations and ethical standards is essential for practical deployment.
CO9	Individual and Team Work	K6: Work effectively as an individual and as a member of multidisciplinary teams. The successful development of the proposed framework requires collaboration among computer scientists, machine learning engineers, ophthalmologists, and healthcare professionals. Each stakeholder contributes	EP5: Team Collaboration. The implementation of a multimodal predictive framework requires continuous communication between AI researchers and medical experts to validate prediction quality and interpretability. Clinical feedback is necessary for selecting meaningful evaluation metrics,

		specialized knowledge during dataset preparation, model development, clinical validation, and system evaluation. Effective teamwork ensures that both technical performance and clinical applicability are addressed throughout the development process.	validating prediction reliability, and ensuring that the framework addresses real-world diagnostic challenges. Collaborative development also supports future improvements and practical deployment in healthcare environments.
CO10	Communication	K3: Communicate engineering concepts effectively through technical reports, system documentation, visualizations, and experimental analysis. The proposed framework presents prediction confidence, agreement, uncertainty, and final diagnostic decisions in an organized and understandable manner. Clear communication enables both technical researchers and healthcare professionals to interpret the prediction results effectively.	EP3: Communication. AI-assisted diagnostic systems should communicate prediction results in a transparent and clinically meaningful way. Rather than presenting only a prediction probability, the proposed framework also provides confidence, agreement, and uncertainty information to improve interpretability. Proper visualization of experimental results, confusion matrices, ROC curves, and performance metrics further enhances communication between researchers and clinicians while supporting evidence-based medical decision-making.
C011	Project Management and Lifelong Learning	K6: Apply project management principles and continuously update engineering knowledge throughout the project lifecycle. The development of CAU-Fusion requires continuous learning in multimodal learning, predictive fusion, explainable AI, uncertainty estimation, and trustworthy medical AI. Proper planning, task scheduling, dataset management, model evaluation, and documentation are essential for successful project completion.	EP7: Sustainable Engineering Practice. The framework should be designed for scalability, maintainability, and future extensibility. As new retinal imaging datasets, clinical variables, and deep learning architectures become available, the framework should be adaptable without major redesign. Continuous model monitoring, periodic retraining, and performance validation are necessary to maintain diagnostic reliability over time. The framework should also support future integration with hospital information systems and clinical decision support platforms while ensuring long-term sustainability and responsible AI deployment.